

Statistical Properties of the Sharpe Ratio Estimator

A Robustness Analysis under Various Assumptions on Returns

Master's Thesis in Banking and Quantitative Finance



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Abstract

This work investigates the finite-sample and asymptotic properties of the Sharpe ratio estimator under progressively realistic return processes, ranging from IID Normal to AR(1)-GARCH(1,1). Combining the method of moments with the delta method, we derive closed-form expressions for the asymptotic variance under each specification. We assess the coverage of asymptotic confidence intervals via Monte Carlo simulation, develop analytical corrections for finite-sample bias, and apply the framework to real financial returns. Complementary analyses cover parameter sensitivity, alternative metrics, and the joint asymptotic distribution of the Sharpe ratio with Value-at-Risk.¹

Keywords: Sharpe ratio · asymptotic variance · AR(1)-GARCH(1,1) · finite-sample bias · model misspecification.

¹Results can be replicated using the code available at https://github.com/Fabro2701/sharpe_ratio_TFM

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1 Introduction

The Sharpe ratio (SR), popularized by Sharpe, 1966, is one of the most widely used tools to measure the performance of an investment. It is defined as the ratio between the expected excess return of an asset or portfolio and its volatility, and offers a simple way to compare strategies with different risk profiles. While the ratio itself is a population quantity, in any practical application it must be computed from a finite sample of returns. The resulting estimator inherits the statistical properties of the underlying return process, so understanding its behaviour is a basic prerequisite for any rigorous use of the metric.

A natural starting point is to assume that returns are independent and identically distributed (IID) Normal, since this assumption leads to simple closed-form results (Lo, 2002). Real financial returns, however, depart from this idealization in well-documented ways. Cont, 2001 reviews a set of stylized facts that appear consistently across assets and markets: heavy tails, asymmetry, volatility clustering, and autocorrelation in low frequencies. These features motivate the use of richer models for the return process and, consequently, a careful study of how the Sharpe ratio estimator behaves under each of them, and what are the risk of missing one of them, a misspecified model.

The existing literature on the statistical properties of the Sharpe ratio estimator can be grouped along three lines: the asymptotic variance, the coverage of asymptotic confidence intervals under correct and incorrect specifications, and the finite-sample bias.

For the asymptotic variance under IID Normal returns, the classical reference is Jobson and Korkie, 1981, with a later and widely cited treatment in Lo, 2002. Mertens, 2002 extends the result to general IID returns by including skewness and kurtosis. Moving beyond the IID setting, Opdyke, 2007 obtains a formula valid under stationarity and ergodicity, while Bao and Ullah, 2006 works with a general covariance structure for the returns. More recent works focus on specific dynamics: López de Prado et al., 2025 treats AR(1) returns; De Capitani, 2012 and Schmid and Schmidt, 2010 cover the GARCH(1,1) case with symmetric innovations; and López de Prado et al., 2026 extends the GARCH(1,1) result to general innovations.

Coverage and misspecification have been studied mainly through simulation. Opdyke, 2007 compares confidence intervals across combinations of SR level, skewness, and kurtosis. López de Prado et al., 2025 carries out an analogous exercise under AR(1) returns, varying also the autocorrelation parameter, and reports situations in which the nominal coverage is not achieved.

The finite-sample bias has received comparatively less attention. Jobson and Korkie, 1981 provides a second-order approximation under IID Normal returns and Miller and Gehr, 1978 gives the corresponding exact expression. Bao and Ullah, 2006 extends the analysis to a general covariance structure of the returns. To the best of our knowledge, no equivalent results are available for other models.

The document is organized as follows. **Section 2** introduces a unified methodology to derive the Sharpe ratio estimator’s asymptotic variance under a wide class of return processes, using the method of moments, the delta method, and asymptotic results for α -mixing sequences. Applying this, we obtain a semi-closed-form expression for the variance under AR(1)-GARCH(1,1) returns; the word “semi” reflects the fact that the odd moments of the conditional variance lack a known analytical solution and are approximated with a Taylor expansion. **Section 3** examines, through Monte Carlo simulation, the coverage of the asymptotic confidence intervals built from these variance formulas under a wide range of configurations, identifying scenarios in which coverage degrades, either because of misspecification or because some regularity conditions fail. **Section 5** applies the variance formulas to real financial data, using standard statistical tests to choose an appropriate model within the family considered in this work. **Section 4** addresses the bias of the estimator in small samples, extending the existing results to all the specifications considered here. Finally, **Section 6** explores two extensions: a skewness-adjusted version of the Sharpe ratio and the joint asymptotic distribution of the Sharpe ratio with the Value-at-Risk.

2 Asymptotic Variance

The Sharpe ratio is a population quantity, but in practice it is reported from a finite sample. The point estimate $\widehat{SR}$ therefore carries sampling error, and any use of it (building a confidence interval, testing significance, ranking strategies) rests on how precisely the estimator tracks its target. That precision is encoded in the asymptotic variance of $\widehat{SR}$, the quantity that governs the spread of $\sqrt{T}(\widehat{SR} - SR)$ as the sample grows. Its size is not an intrinsic property of the estimator: it depends on the model assumed for the return process. The same point estimate computed on IID Gaussian returns and on a heavy-tailed, autocorrelated, conditionally heteroskedastic series will carry markedly different statistical weight, since persistence in the levels and clustering in the volatility both change the effective dispersion of the sample moments. This section derives the asymptotic variance of $\widehat{SR}$ under the four return models considered in this work, namely IID, AR(1), GARCH(1,1), and AR(1)-GARCH(1,1), within a single inferential framework based on the Method of Moments and the delta method. The framework concentrates the effect of the assumed dynamics in one object, the long-run variance matrix of the moment conditions, so that each model becomes a particularisation of a single general formula.

2.1 Methodology

The inference strategy combines the *Method of Moments* (MM) with the *delta method*. The MM yields the asymptotic distribution of the parameter vector $\hat{\theta}$, and the delta method propagates that distribution to the nonlinear functional of interest, the Sharpe ratio.

Process and parameters. Let $\{r_t\}_{t=1}^T$ be a stationary process with finite first two moments,

$$E[r_t] = \mu, \quad \text{Var}(r_t) = \sigma^2.$$

The parameter vector to be estimated is $\theta = (\mu, \sigma^2)^\top$. Because the Sharpe ratio depends on the return distribution only through its first two moments, this minimal parameterisation is sufficient to characterise $\widehat{SR}$, irrespective of the higher-order dynamics of the underlying process. Those higher-order features (autocorrelation, conditional heteroskedasticity, heavy tails) re-enter the analysis through the long-run variance of the moment conditions defined below.

Moment conditions. The moment conditions are

$$m(r_t, \theta) = \begin{pmatrix} r_t - \mu \\ (r_t - \mu)^2 - \sigma^2 \end{pmatrix}, \quad (1)$$

which satisfy, by construction, $E[m(r_t, \theta)] = \mathbf{0}$ at the true parameter value. The first condition identifies the mean and the second identifies the variance. The system is exactly identified, since the dimensions of m and θ coincide; the resulting MM estimator coincides with the sample mean and the sample variance with denominator T .

Long-run variance. Write $m_t \equiv m(r_t, \theta)$ and define the autocovariance function of the moment conditions at lag h as

$$\Gamma(h) = E[m_t m_{t-h}^\top], \quad h \in \mathbb{Z}. \quad (2)$$

Let $\bar{m}_T(\theta) \equiv T^{-1} \sum_{t=1}^T m(r_t, \theta)$ denote the sample average of the moment conditions. The variance of the normalised sum $\sqrt{T} \bar{m}_T(\theta)$ admits the decomposition

$$\begin{aligned} \text{Var}\left(\sqrt{T} \bar{m}_T(\theta)\right) &= \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T E[m_t m_s^\top] = \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T \Gamma(t-s) \\ &= \sum_{h=-(T-1)}^{T-1} \left(1 - \frac{|h|}{T}\right) \Gamma(h) \xrightarrow{T \rightarrow \infty} S \equiv \sum_{h=-\infty}^{\infty} \Gamma(h), \end{aligned} \quad (3)$$

where S is the *long-run variance* (equivalently, the spectral density of $\{m_t\}$ evaluated at frequency zero). The second equality is obtained by grouping the pairs (t, s) with common lag $h = t - s$, which occur exactly $T - |h|$ times. The convergence to S is justified by dominated convergence once $\sum_h \|\Gamma(h)\| < \infty$, a condition examined below as part of the regularity assumptions.

Structure of $\Gamma(h)$. Substituting (1) into (2) and introducing the centred variable $x_t = r_t - \mu$ gives

$$\Gamma(h) = E \left[\begin{pmatrix} x_t \\ x_t^2 - \sigma^2 \end{pmatrix} \begin{pmatrix} x_{t-h} & x_{t-h}^2 - \sigma^2 \end{pmatrix} \right] = \begin{pmatrix} \text{Cov}(x_t, x_{t-h}) & \text{Cov}(x_t, x_{t-h}^2) \\ \text{Cov}(x_t^2, x_{t-h}) & \text{Cov}(x_t^2, x_{t-h}^2) \end{pmatrix}, \quad (4)$$

where the centring $E[x_t] = 0$ allows every entry to be written directly as a covariance. The four entries have a clear interpretation: $\Gamma(h)_{11}$ is the autocovariance of the levels at lag h , $\Gamma(h)_{22}$ is the autocovariance of the centred squared returns at lag h , and the off-diagonal entries $\Gamma(h)_{12}$, $\Gamma(h)_{21}$ are the cross-covariances between levels and squared centred returns.

Regularity conditions and CLT for the moment conditions. To establish the asymptotic normality of the estimator in the presence of temporal dependence (such as AR dynamics) and conditional heteroskedasticity (such as GARCH effects), the IID assumption must be dropped and a Central Limit Theorem (CLT) for dependent and heterogeneous processes invoked. We rely on the asymptotic framework of White, 2001, Section 5.4, Theorem 5.20.

Since m_t is a fixed function of the observed returns, standard econometric practice imposes the regularity conditions directly on the primitive process $\{r_t\}$. Three joint conditions are required:

1. **Strong mixing (α -mixing) with controlled decay rate.** To accommodate temporal dependence, the return process $\{r_t\}_{t \in \mathbb{Z}}$ is assumed to be *strongly mixing* (or α -mixing) in the sense of White, 2001, Section 3.42. Given the σ -algebras

$$\mathcal{F}_{-\infty}^t = \sigma(\dots, r_{t-1}, r_t) \quad \text{and} \quad \mathcal{F}_{t+h}^\infty = \sigma(r_{t+h}, r_{t+h+1}, \dots),$$

the *mixing coefficient* is defined as

$$\alpha(h) = \sup_{t \in \mathbb{Z}} \sup_{\substack{A \in \mathcal{F}_{-\infty}^t \\ B \in \mathcal{F}_{t+h}^\infty}} |\Pr(A \cap B) - \Pr(A) \Pr(B)|, \quad h \geq 0. \quad (5)$$

The process is α -mixing whenever $\alpha(h) \rightarrow 0$ as $h \rightarrow \infty$, formalising the idea that events sufficiently separated in time become asymptotically independent. Following the Wooldridge-White Theorem White, 2001, the mixing coefficients are required to be of size $-r/(r-2)$ for some $r > 2$, i.e.

$$\alpha(h) = O\left(h^{-(\frac{r}{r-2} + \varepsilon)}\right) \quad \text{for some } \varepsilon > 0. \quad (6)$$

This rate implies the summability condition

$$\sum_{h=0}^{\infty} \alpha(h)^{1-2/r} < \infty. \quad (7)$$

To see it, raise (6) to the exponent $1 - 2/r = (r-2)/r$, obtaining

$$\alpha(h)^{1-2/r} = O\left(h^{-(\frac{r}{r-2} + \varepsilon) \frac{r-2}{r}}\right) = O\left(h^{-(1+\delta)}\right), \quad \delta := \varepsilon \frac{r-2}{r} > 0.$$

Since $\sum_{h=1}^{\infty} h^{-(1+\delta)} < \infty$ for any $\delta > 0$, condition (7) follows by comparison.

This requirement establishes a strict *trade-off* between the memory of the process and the tail thickness of its marginal distribution: because the exponent $1 - 2/r \in (0, 1)$ is increasing in r , a process with very heavy tails (finite moment of order r close to two) demands that $\alpha(h)$ decay to zero extremely fast for the series to be summable; conversely, when the tails are thin ($r \rightarrow \infty$, exponent approaching one), considerably more persistent temporal dependence is admissible.

2. **Existence of moments of order r .** Consistent with the parameter introduced in the previous condition, the absolute moment of order r of the moment conditions is required to be bounded,

$$E[\|m_t\|^r] < \Delta < \infty, \quad (8)$$

for the same $r > 2$. Given the quadratic form in (1), this is equivalent to requiring the existence of the moment of order $2r$ of the returns, $E[|r_t|^{2r}] < \infty$. For financial distributions with heavy tails this is a restrictive condition that may invalidate both the normal approximation for $\widehat{SR}$ and the canonical $\sqrt{T}$ scaling; the implications of this failure are examined in Section 3.

3. **Long-run variance strictly positive and finite.** White, 2001, Lemma 6.16 shows that, for an α -mixing process, $\alpha(h) \rightarrow 0$ implies $\|\Gamma(h)\| \rightarrow 0$ as $h \rightarrow \infty$, and combined with the moment bound above it yields

$$\sum_{h=-\infty}^{\infty} \|\Gamma(h)\| < \infty, \quad (9)$$

so that the matrix S in (3) is well defined under dependence and heteroskedasticity. As shown in White, 2001, Theorem 6.20, an estimator of this variance is consistent.

Under these three conditions, the CLT for dependent processes applies, so that

$$\sqrt{T} \bar{m}_T(\theta) = \frac{1}{\sqrt{T}} \sum_{t=1}^T m_t \xrightarrow{d} \mathcal{N}(\mathbf{0}, S). \quad (10)$$

Expected Jacobian. The expected Jacobian of the moment conditions, evaluated at the true parameter θ , is

$$H = E \left[\frac{\partial m(r_t, \theta)}{\partial \theta^\top} \right] = E \begin{pmatrix} -1 & 0 \\ -2(r_t - \mu) & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I_2, \quad (11)$$

where $E[r_t - \mu] = 0$ eliminates the lower-left entry.

Asymptotic distribution of the MM estimator. The MM estimator $\hat{\theta}$ is defined as the solution of

$$\bar{m}_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T m(r_t, \hat{\theta}) = \mathbf{0}. \quad (12)$$

A first-order Taylor expansion of $\bar{m}_T(\cdot)$ around the true parameter θ gives

$$\mathbf{0} = \bar{m}_T(\hat{\theta}) \approx \bar{m}_T(\theta) + H(\hat{\theta} - \theta), \quad (13)$$

Solving for $\hat{\theta}$ and rescaling,²

$$\sqrt{T}(\hat{\theta} - \theta) \approx -H^{-1} \sqrt{T} \bar{m}_T(\theta). \quad (14)$$

Combining this linearisation with the CLT in (10),

$$\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, H^{-1} S (H^{-1})^\top). \quad (15)$$

Substituting $H = -I_2$ from (11) simplifies the sandwich to the long-run variance itself,

$$\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, S). \quad (16)$$

The whole structure of temporal dependence and conditional heteroskedasticity of the process is encoded in the matrix S through the sum of autocovariances of the moment conditions.

²The matrix H is square because the system is exactly identified ($\dim m = \dim \theta = 2$). The inversion step requires H to be nonsingular at the true parameter.

Delta method and asymptotic variance of the Sharpe ratio. The Sharpe ratio is a smooth function of θ ,

$$g(\theta) = \frac{\mu}{\sigma} = \frac{\mu}{\sqrt{\sigma^2}},$$

with gradient

$$\nabla_{\theta} g = \begin{pmatrix} \frac{\partial g}{\partial \mu} \\ \frac{\partial g}{\partial \sigma^2} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sigma} \\ -\frac{\mu}{2\sigma^3} \end{pmatrix}. \quad (17)$$

Applying the delta method to (16),

$$\sqrt{T}(\widehat{SR} - SR) \xrightarrow{d} \mathcal{N}(0, V_{\widehat{SR}}), \quad V_{\widehat{SR}} = (\nabla_{\theta} g)^{\top} S \nabla_{\theta} g. \quad (18)$$

Writing S in component form,

$$S = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix}, \quad S_{ij} = \sum_{h=-\infty}^{\infty} \Gamma(h)_{ij}, \quad (19)$$

note that S is *symmetric*. Indeed, $\Gamma(h)^{\top} = \Gamma(-h)$ follows directly from the definition (2), so that

$$S_{21} = \sum_{h=-\infty}^{\infty} \Gamma(h)_{21} = \sum_{h=-\infty}^{\infty} \Gamma(-h)_{12} = \sum_{h=-\infty}^{\infty} \Gamma(h)_{12} = S_{12},$$

where the last step uses the change of index $h \mapsto -h$, which leaves a sum over all integers unchanged. Therefore $S_{12} = S_{21}$.

Substituting into (18) and computing the row-times-matrix product first,

$$(\nabla_{\theta} g)^{\top} S = \left(\frac{S_{11}}{\sigma} - \frac{\mu S_{21}}{2\sigma^3}, \quad \frac{S_{12}}{\sigma} - \frac{\mu S_{22}}{2\sigma^3} \right). \quad (20)$$

Multiplying on the right by $\nabla_{\theta} g$ and using $S_{12} = S_{21}$,

$$\begin{aligned} V_{\widehat{SR}} &= \left(\frac{S_{11}}{\sigma} - \frac{\mu S_{12}}{2\sigma^3} \right) \frac{1}{\sigma} + \left(\frac{S_{12}}{\sigma} - \frac{\mu S_{22}}{2\sigma^3} \right) \left(-\frac{\mu}{2\sigma^3} \right) \\ &= \frac{S_{11}}{\sigma^2} - \frac{\mu S_{12}}{2\sigma^4} - \frac{\mu S_{12}}{2\sigma^4} + \frac{\mu^2 S_{22}}{4\sigma^6} \\ &= \boxed{\frac{S_{11}}{\sigma^2} - \frac{\mu S_{12}}{\sigma^4} + \frac{\mu^2 S_{22}}{4\sigma^6}}. \end{aligned} \quad (21)$$

Expression (21) is the general formula for the asymptotic variance of the Sharpe ratio estimator under any stationary process satisfying the regularity conditions stated above. Particularising S to a specific model yields the asymptotic variance for that model, as carried out in Section 2.2.

2.2 Model Specifications and Asymptotic Variances

Under the regularity conditions of Section 2.1, the Sharpe ratio estimator admits the asymptotic distribution

$$\sqrt{T}(\widehat{SR} - SR) \xrightarrow{d} \mathcal{N}(0, V_{\widehat{SR}}), \quad (22)$$

where $V_{\widehat{SR}}$ is determined by the higher moments of the return distribution and by its temporal dependence structure, packaged jointly in the long-run variance matrix S of the moment conditions through the general formula (21). Specialising S to each model considered in this work yields the closed-form expressions reported in Table 1. The first five rows reproduce results already available

in the literature. The AR(1)-GARCH(1,1) result is derived in Appendix B and constitutes the principal analytical contribution of this section.

Table 1: Return models and asymptotic variance formulas.

Model	Return Equation	Asymptotic Variance ($V_{\widehat{SR}}$)
IID Normal ^a	$r_t \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$	$1 + \frac{1}{2}SR^2$
IID Non-Normal ^b	$r_t \stackrel{\text{iid}}{\sim} (\mu, \sigma^2)$	$1 - \gamma SR + \frac{\kappa-1}{4}SR^2$
AR(1) Normal	$\begin{cases} r_t = c + \phi r_{t-1} + \epsilon_t \\ \epsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma_\epsilon^2) \end{cases}$	$\frac{1+\phi}{1-\phi} + \frac{1}{2} \frac{1+\phi^2}{1-\phi^2} SR^2$
AR(1) Non-Normal ^c	$\begin{cases} r_t = c + \phi r_{t-1} + \epsilon_t \\ \epsilon_t \stackrel{\text{iid}}{\sim} (0, \sigma_\epsilon^2) \end{cases}$	$\frac{1+\phi}{1-\phi} - \frac{1+\phi+\phi^2}{1-\phi^2} \gamma SR + \frac{1+\phi^2}{1-\phi^2} \frac{\kappa-1}{4} SR^2$
GARCH(1,1) ^d	$\begin{cases} r_t = \mu + u_t \\ u_t = \sigma_t \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} (0, 1) \\ \sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2 \end{cases}$	$1 - \frac{1-\beta}{1-\alpha-\beta} \gamma SR + \frac{\kappa-1}{4} \frac{(1-\beta)^2(1+\alpha+\beta)}{(1-\alpha-\beta)(1-2\alpha\beta-\beta^2)} SR^2$
AR(1)-GARCH(1,1) sym	$\begin{cases} r_t = c + \phi r_{t-1} + u_t \\ u_t = \sigma_t \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} (0, 1) \\ E[\epsilon_t^3] = 0 \\ \sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2 \end{cases}$	$\left\{ \frac{1+\phi}{1-\phi} + \frac{1}{4} \frac{1}{1-\phi^2} \left[4\phi^2 + \frac{K}{1+A} \left(\frac{2}{3} A + \frac{(1-\beta)^2(1+\alpha+\beta)}{(1-\alpha-\beta)(1-2\alpha\beta-\beta^2)} \right) \right] \right\} SR^2$ $A = 6\phi^2 \alpha \frac{1-\alpha-\beta}{1-2\alpha\beta-\beta^2} \frac{1}{1-\phi^2(\alpha+\beta)}$ $K = (1+\phi^2)\kappa - 5\phi^2 - 1$

Note: $\gamma = E[(r_t - \mu)^3]/\sigma^3$ denotes the skewness and $\kappa = E[(r_t - \mu)^4]/\sigma^4$ the kurtosis of the marginal return distribution. The symbol $SR = \mu/\sigma$ denotes the population Sharpe ratio.

^a (Lo, 2002)

^b (Mertens, 2002)

^c (López de Prado et al., 2025)

^d (López de Prado et al., 2026)

A side-by-side comparison of the simpler rows of Table 1 against the Gaussian benchmark $V_{\widehat{SR}}^{\text{IID N}} = 1 + \frac{1}{2}SR^2$ admits a clean qualitative reading. In the IID non-Gaussian model, $V_{\widehat{SR}}$ is monotonically increasing in the kurtosis κ and decreasing in the skewness γ whenever $SR > 0$: heavier-than-Gaussian tails always inflate the variance of $\widehat{SR}$, and negative skewness inflates it further. The same direction of effects propagates to the AR(1) non-Gaussian row, with ϕ entering as a multiplicative amplifier through the factor $(1+\phi)/(1-\phi)$ on the constant term and through related rational functions of ϕ on the coefficients of γSR and SR^2 . For the GARCH(1,1) and AR(1)-GARCH(1,1) specifications this kind of by-inspection reading is no longer possible: the volatility-dynamics parameters (α, β) enter both the constant and the SR -coefficients in ways that interact with (γ, κ) and, in the joint case, with ϕ . The numerical sensitivity analysis of Section 2.3 maps these interactions across the relevant region of the parameter space.

2.3 Sensitivity of the variance to model parameters

This section examines how the asymptotic variance of the Sharpe ratio estimator under AR(1)-GARCH(1,1) returns, $V_{\widehat{SR}}$, responds to changes in the model parameters and in the higher-order moments collected in θ . Two complementary exercises are combined: a one-at-a-time (OAT) traversal, which measures the local sensitivity around a chosen base point through numerical evaluation of the variance formula, and an axis-stratified Monte Carlo (MC) experiment, whose output is independent of that base point.

A parameter vector θ is called *admissible* when every component lies inside the region where the model is well defined. The constraints are AR(1) stationarity ($|\phi| < 1$), GARCH(1,1) stationarity ($\alpha + \beta < 1$), existence of the fourth moment of the return process ($\kappa_\epsilon \alpha^2 + 2\alpha\beta + \beta^2 < 1$, with κ_ϵ the kurtosis of the innovation), a kurtosis of the returns no smaller than the Gaussian benchmark

($\kappa_r \geq 3$, which is automatically satisfied once the GARCH dynamics are active), and a positive Sharpe ratio ($SR > 0$). Every point that violates any of these constraints is discarded from the analysis.

OAT analysis. Each parameter θ_i is varied along its admissible interval while the remaining components are held at the base point

$$\bar{\theta} = (\overline{SR}, \bar{\phi}, \bar{\alpha}, \bar{\beta}, \overline{\kappa_r - 3}, \bar{s_r}) = (0.5, 0.2, 0.08, 0.87, 5.0, 0.0).$$

Figure 1 reports the resulting curves: dashed lines mark the base value on each axis and the red dot marks $V_{\widehat{SR}}(\bar{\theta}) \approx 4.36$.

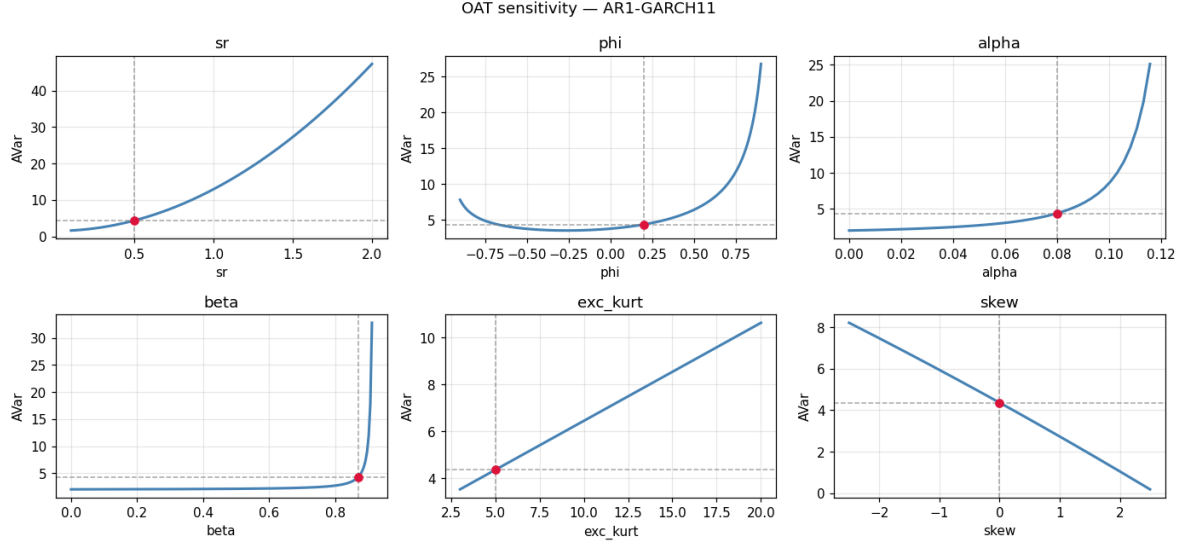


Figure 1: OAT traversal: each panel varies a single parameter while the others remain at the base point. Dashed lines indicate base values. The red dot marks $V_{\widehat{SR}}(\bar{\theta})$.

The local sensitivity profiles are heterogeneous:

- SR : monotonically increasing and convex, in line with the quadratic term $\frac{1}{4}SR^2S_{22}$ of the expansion. $V_{\widehat{SR}}$ moves from ≈ 1.6 at $SR = 0$ to ≈ 47.3 at $SR = 2$.
- ϕ : asymmetric U-shape with an interior minimum near $\phi \approx -0.25$. The right branch rises sharply as $\phi \rightarrow 1$ and diverges at the AR stationarity boundary $|\phi| = 1$. The left branch also rises, but more mildly, because of the interaction of ϕ with the third and fourth moments of the process. The base value $\phi = 0.2$ lies close to the minimum.
- α : with $\beta = 0.87$ held fixed, the admissible range is restricted to $\alpha < 1 - \beta = 0.13$, and the fourth-moment constraint narrows it further. $V_{\widehat{SR}}$ grows exponentially and diverges as the IGARCH boundary $\alpha + \beta = 1$ is approached.
- β : essentially flat over $\beta \in [0, 0.8]$ (variation below 5%) and explosive beyond $\beta \approx 0.9$. The base value $\beta = 0.87$ already sits at the edge of the explosive regime.
- κ_r : monotonically increasing and approximately linear in $V_{\widehat{SR}}$.
- s_r : monotonically decreasing and approximately linear in $V_{\widehat{SR}}$. The contribution of the quadratic term in s_r^2 is negligible at the chosen base point and is therefore not visible in the OAT curve.

The divergences seen on the α and β axes are local artefacts of holding the other GARCH coefficient close to the non-stationarity boundary. Reading them as global statements about parameter sensitivity is misleading, which motivates the second exercise.

Axis-stratified Monte Carlo. For each parameter θ_i , $n = 4000$ samples are generated in which θ_i is drawn uniformly over its admissible interval and the remaining parameters are drawn uniformly on their respective bounds, then accepted or rejected against the joint admissibility constraints. Stratifying by axis guarantees uniform coverage of θ_i over its full range, which a single rejection scheme on the global box would fail to deliver near the upper end of α , where the admissible slice in (β, κ_r) shrinks. Because every parameter is sampled rather than fixed, the conclusions of this experiment do not depend on the base point chosen for the OAT.

For each axis, the Spearman rank correlation between θ_i and $V_{\widehat{SR}}$ is computed across the stratified sample as a non-parametric measure of monotonic dependence (Table 2); an OLS line is overlaid on the scatter in Figure 2 together with its slope, R^2 , and p -value.

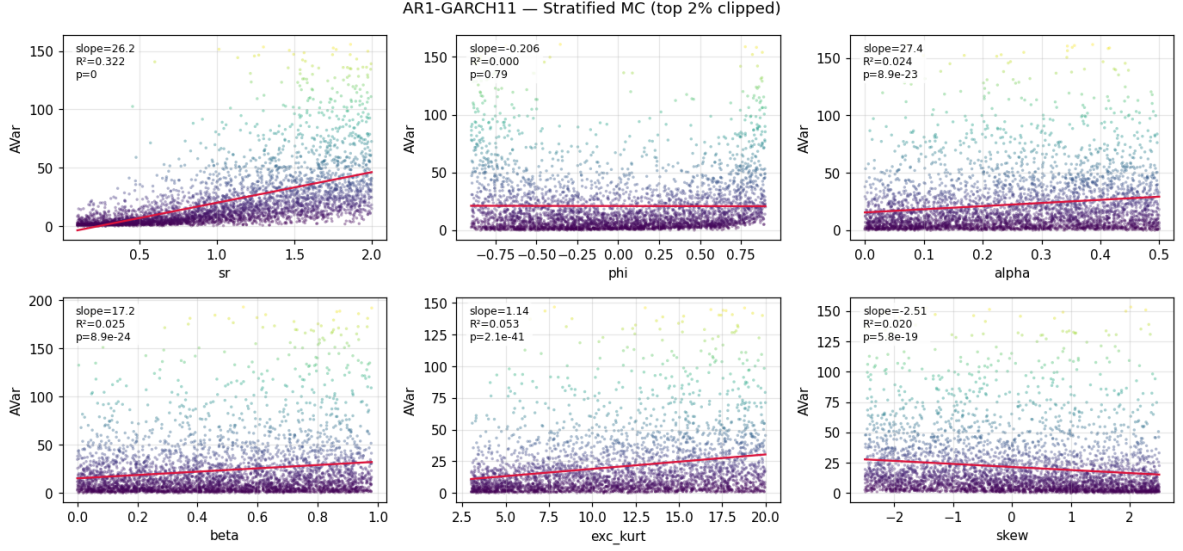


Figure 2: Axis-stratified Monte Carlo ($n = 4000$ per panel). Each cloud is annotated with the slope, R^2 , and p -value of an OLS fit.

Axis	Spearman ρ_S
SR	0.74
ϕ	0.05
α	0.20
β	0.14
κ_r	0.25
s_r	-0.22

Table 2: Spearman rank correlation between each axis parameter and $V_{\widehat{SR}}$ under axis-stratified Monte Carlo sampling.

Joint reading. The comparison of OAT curves and stratified MC clouds yields the following observations:

- SR is the dominant driver of $V_{\widehat{SR}}$ under every metric: the highest rank correlation ($\rho_S = 0.74$), the largest R^2 in the OLS fit of Figure 2, and a clean quadratic profile in the OAT. The dependence is structural rather than boundary-driven.
- ϕ : the scatter cloud traces a horseshoe (interior minimum with rising branches at both extremes) that a linear fit only partially captures. The result is a near-zero OLS slope combined with a negligible R^2 and an almost null rank correlation. This is the only axis for which the slope is not statistically distinguishable from zero.

- α and β display mild rank correlations (0.20 and 0.14, respectively) and small R^2 values. Their effect is strongly non-linear and concentrated near the boundaries: explosions occur at the IGARCH frontier $\alpha + \beta = 1$ for the GARCH coefficients and at $|\phi| \rightarrow 1$ for the AR coefficient, but those regions account for a small fraction of the admissible space, so the average response across feasibility is moderate.
- κ_r is the second most informative factor ($\rho_S = 0.25$). Its effect is approximately linear, monotonic, and consistent across the rest of the parameter space.
- s_r has almost the same magnitude as κ_r ($\rho_S = -0.22$), with a linear, monotonic, and consistent effect across the rest of the parameter space.

Disagreement between the two exercises does not signal a defect of either; it signals that the effect of the parameter in question is non-linear or boundary-driven. In the present model, SR , ϕ , s_r , and κ_r behave as robust drivers under both readings, whereas the variance contribution of α and β depends strongly on proximity to the fourth-moment frontier. And the effect of ϕ seems to be quadratic rather than monotonically.

3 Coverage Analysis

Opdyke, 2007 showed that, under IID returns, the classical formula assuming normality systematically underestimates the variance of the Sharpe ratio estimator in the presence of heavy tails and negative skewness, which translates into confidence intervals that are too narrow. That study, however, is restricted to the IID case and leaves unexamined the joint effect of two important features of financial returns: serial dependence in the mean and conditional heteroscedasticity. López de Prado et al., 2025 make some analysis on the effect of the first order correlation, but lack some interesting features, like the asymmetry around 0 (negative correlation affects different than positive) or the behaviour around the non-stationarity levels.

The aim of this section is to extend those analysis along two complementary axes. First, we evaluate the empirical coverage of the asymptotic interval under the candidate model families described in Section 2.2, we quantify the cost of misspecification: what happens to coverage when the model assumed by the analyst does not match the true data-generating process, i.e. it misses non-normality, persistence in the mean, conditional volatility, and combinations of both, together with the behaviour of coverage under correct specification. Second, more specific configurations (different mean persistences ϕ , different volatility persistences $\alpha + \beta$, and different Sharpe ratio levels). The simulation methodology is described in Appendix A.1.

Why evaluate coverage rather than the variance directly?

A seemingly simpler alternative would be to simulate trajectories and check that the sample variance of $\widehat{SR}$ converges to the value predicted by the asymptotic formula. Coverage is, however, the preferable metric for several reasons.

- **It is the relevant inferential quantity.** What matters to the end user is not that the variance be well estimated *per se*, but that the confidence intervals contain the parameter at the announced frequency. Coverage is the *output*; the variance is only an intermediate step.
- **It is a joint test.** An incorrect empirical coverage can be due to (i) a poorly estimated variance, (ii) bias in the point estimator, or (iii) the finite-sample distribution of $\widehat{SR}$ being far from Normal. Comparing variances detects only the first failure, whereas coverage is sensitive to all three.
- **It has a calibrated target and an associated formal test.** Under the null hypothesis of correct specification, the empirical coverage follow a binomial distribution which gives rise to acceptance bands (See Appendix A.1). Comparing variances, by contrast, requires fixing an arbitrary tolerance.
- **It is comparable across scenarios.** The asymptotic variance changes in magnitude with SR , T and the moments of the DGP; coverage, in contrast, always lies in $[0, 1]$ with the same nominal target, so highly heterogeneous configurations can be compared on a common scale.

3.1 Stylized facts

We now present a series of scenarios designed to isolate, one feature at a time, the impact of each property of the returns on the empirical coverage of the asymptotic interval. The ordering is deliberately cumulative: from the simplest failure of the normality assumption (Scenario 1) to the most complex DGP, in which serial dependence and conditional heteroscedasticity coexist (Scenario 5). Unless stated otherwise, every experiment in this subsection uses $n = 50,000$ Monte Carlo replications, targets a nominal coverage of $1 - \alpha = 0.95$, fixes the population Sharpe ratio at $SR = 0.5$ and the volatility at $\sigma = 1$, and reports the empirical coverage on the grid $T \in \{100, 300, 500, 1,000, 2,000\}$, displayed on a logarithmic horizontal axis. Variance estimates use the theoretical-moments plug-in described in step (3) of the protocol (See Appendix A.1); this

choice isolates the asymptotic formula from the additional noise of estimating θ , which matters in particular for the GARCH-based variance, where parameter estimation is unstable at moderate T . The specific DGP parameters used in each scenario are stated in the corresponding subsubsection.

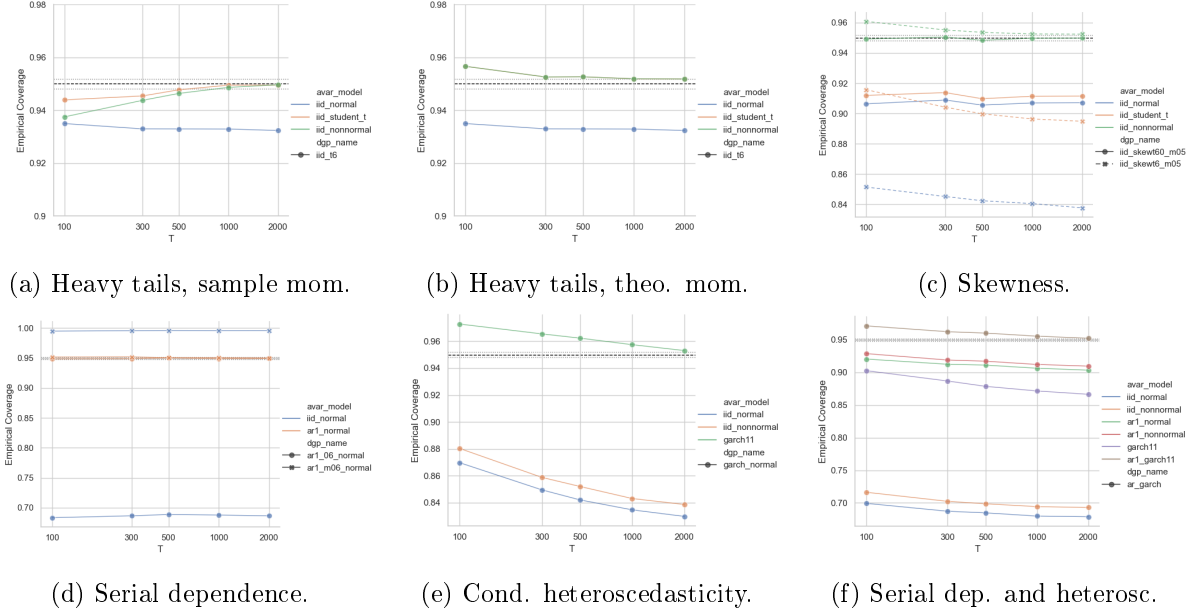


Figure 3: Empirical coverage of the nominal 95% asymptotic confidence interval for the Sharpe ratio estimator as a function of the sample size T (logarithmic axis), across the five stylized-fact scenarios. Each panel reports the coverage attained by the candidate models under the corresponding DGP. The horizontal band around 0.95 is the binomial acceptance region defined in (53). All curves are based on $n = 50,000$ Monte Carlo replications.

3.1.1 Scenario 1: heavy tails

The DGP is IID with Student- t innovations and six degrees of freedom, which produces heavy tails relative to the Normal distribution. Three candidate models are evaluated: IID Normal, IID Student- t and IID general. Figure 3a shows that the Normal model underestimates the variance of the estimator and converges from below to a coverage strictly lower than the nominal level. The IID Student- t and IID general models, in contrast, attain the nominal level once T is large enough to drive the asymptotic approximation into its regime of validity.

Figure 3b reports the same configuration evaluated with theoretical moments. The IID Student- t and IID general curves now overlap exactly, whereas with sample moments (panel a) they differ slightly: the IID general model estimates a sample skewness different from zero owing to estimation noise, even though the true skewness of the t distribution is zero, and this spurious term feeds the variance formula and introduces a small discrepancy. The Normal model, of course, remains biased in both panels, since the gap is intrinsic to the formula rather than to the estimation of its inputs.

3.1.2 Scenario 2: skewness

We now use two skew- t DGPs that share the same moderate negative skewness but differ in tail thickness: the first has 60 degrees of freedom, which keeps the kurtosis close to the Normal value, and the second has 6 degrees of freedom, which produces heavy tails. Three candidate models are evaluated: IID Normal, IID Student- t and IID general. Figure 3c shows that only the IID general model, which carries the skewness parameter inside its variance formula, converges to the nominal coverage in both cases. The Normal and Student- t models, by assuming symmetry, fail to correct the bias induced by the asymmetry of the DGP. Their performance is noticeably better under

the first DGP (light tails) than under the second (heavy tails), which indicates that the variance contribution of skewness is amplified when the tails are also heavy: holding the skewness fixed, increasing the kurtosis makes the symmetry assumption costlier.

3.1.3 Scenario 3: serial dependence (AR(1))

We now consider AR(1) processes with Normal innovations and two opposite values of the autoregressive coefficient, $\rho = 0.6$ and $\rho = -0.6$. Two candidate models are evaluated: IID Normal and AR(1) Normal. Figure 3d shows a clear pattern of **systematic under-coverage** when the IID Normal model is faced with positive persistence ($\rho = 0.6$), because positive autocorrelation inflates the long-run variance of the returns and the IID formula ignores this contribution.

The result is highly **asymmetric in the sign of the autocorrelation**. The IID Normal model copes much better with negative persistence ($\rho = -0.6$), where it reaches coverages close to 0.99, than with positive persistence, where coverage drops to roughly 0.69. The AR(1) Normal model, which incorporates the autoregressive coefficient into the variance formula, attains essentially nominal coverage in both cases.

3.1.4 Scenario 4: conditional volatility (GARCH)

We turn to a GARCH(1,1) DGP with Normal innovations and ask whether capturing the unconditional kurtosis through the IID general model is enough to achieve coverage. The three candidate models are IID Normal, IID general and GARCH(1,1). Figure 3e shows that the IID general model improves coverage consistently relative to the IID Normal one, but remains far below the GARCH-specific model, which attains the nominal level. The intuition is that the GARCH dynamics induce serial dependence in the squared returns that the IID general formula cannot represent, even after correcting for the heavy unconditional tails.

3.1.5 Scenario 5: heteroscedasticity and serial dependence

The last scenario combines the two effects in an AR(1)-GARCH(1,1) process, with autocorrelation in the mean and conditional heteroscedasticity in the variance. Six candidate models are evaluated: IID Normal, IID general, AR(1) Normal, AR(1) general, GARCH(1,1) and AR(1)-GARCH(1,1) with symmetric innovations. Figure 3f shows that only the AR(1)-GARCH(1,1) specification, which is the correctly-specified one in the candidate list, attains the nominal coverage. The other five models display systematic under-coverage, with a clear hierarchy: the AR(1) variants reach coverages close to 0.90 thanks to their partial capture of the mean dependence, the pure GARCH model captures the conditional heteroscedasticity but leaves the mean persistence unmodelled, and the IID alternatives perform worst. The exercise motivates the derivation of the AR(1)-GARCH(1,1) asymptotic variance carried out in Section 2, which is the only formula in this menu that closes the coverage gap when both stylized facts are present in the data.

3.2 Additional scenarios

The five stylized-fact scenarios of the previous subsection explore the role of each return feature in isolation, but always at a single configuration of the corresponding parameter. We now turn to a second battery of experiments that explore the asymptotic interval beyond those configurations along four additional dimensions: the speed of convergence to the nominal coverage as the persistence of a correctly-specified model varies (Scenarios a1 and a2); the impact of the population Sharpe ratio itself (Scenario a3); the breakdown of the asymptotic approximation when the fourth moment of the returns does not exist (Scenario a4); and the finite-sample cost of overparameterizing the variance formula (Scenario a5). All experiments share the simulation protocol of the previous subsection: $n = 50,000$ Monte Carlo replications, nominal level $1 - \alpha = 0.95$, volatility $\sigma = 1$ and population Sharpe ratio $SR = 0.5$ unless it is the variable being swept. The sample size sweeps $T \in \{100, 300, 500, 1,000, 2,000\}$ on a logarithmic axis, except in Scenario a3, where T is fixed at

100 and the Sharpe ratio is swept instead. Variance estimates use sample moments throughout, with the sole exception of Scenario a2, where the GARCH parameter estimation is unstable at moderate T and we report theoretical moments.

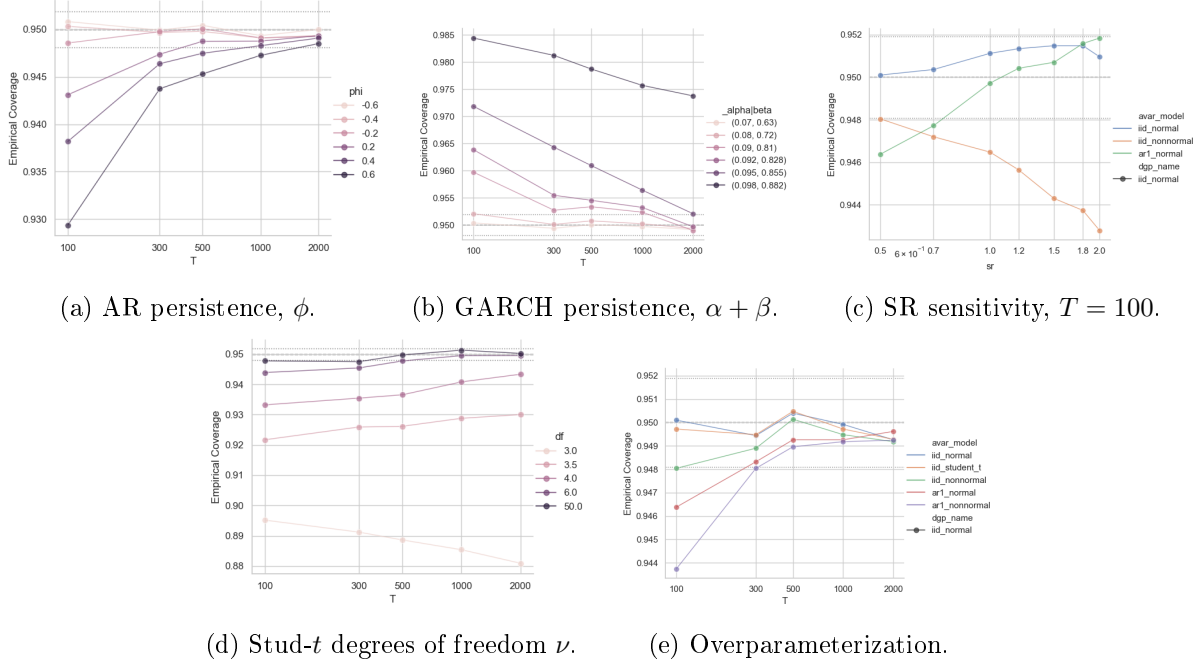


Figure 4: Empirical coverage of the nominal 95% asymptotic confidence interval for the Sharpe ratio estimator across the five additional scenarios. In panels (a), (b) and (d) each curve corresponds to a different value of the parameter being swept under a correctly-specified candidate model; in panels (c) and (e) each curve corresponds to a different candidate model with the DGP held fixed at IID Normal. The horizontal band around 0.95 is the binomial acceptance region defined in (53), and all curves are based on $n = 50,000$ Monte Carlo replications.

3.2.1 Scenario a1: persistence in the mean

The DGP is an AR(1) with Normal innovations and the candidate model is the correctly-specified AR(1) Normal. The autoregressive coefficient sweeps $\phi \in \{-0.6, -0.4, -0.2, 0.2, 0.4, 0.6\}$, which lets us trace how the speed of convergence to nominal coverage depends on the strength and the sign of the serial dependence. Figure 4a shows that every curve converges to the nominal level as T grows, as expected from a correctly-specified model, but the speed of convergence deteriorates with ϕ : for small persistences ($\phi = 0.2$) the coverage enters the acceptance band already at $T = 500$, while for $\phi = 0.6$ the convergence is visibly slower. The sign of ϕ matters as well, with positive persistence producing the slower convergence at any given level of $|\phi|$, consistent with the asymmetry observed in Scenario 3 of the previous subsection.

3.2.2 Scenario a2: persistence in the volatility

The DGP is a GARCH(1,1) with Normal innovations and the candidate model is the correctly-specified GARCH(1,1) variance formula. The persistence $\alpha + \beta \in \{0.7, 0.8, 0.9, 0.92, 0.95, 0.98\}$ and, at each level. The corresponding tail indices, derived in Section A.2, are approximately $\{26.46, 20.44, 14.08, 12.49, 9.64, 5.74\}$: more persistent volatility implies thicker unconditional tails, even with Normal innovations. Figure 4b shows that the convergence of the coverage to the nominal level is monotonically slower as persistence grows. For $\alpha + \beta = 0.7$ the nominal level is essentially reached at the smaller sample sizes; for $\alpha + \beta = 0.98$ the coverage remains visibly below target even at $T = 2,000$. The tail index offers a clean reading of the pattern: the GARCH

variance formula requires the fourth moment of the returns to be finite, and while every configuration evaluated here satisfies that condition (every tail index exceeds 4), the closer the index gets to 4 the more the higher-order sample moments behave erratically, slowing the convergence of the asymptotic approximation.

3.2.3 Scenario a3: sensitivity to the Sharpe ratio level

We now fix the sample size at $T = 100$ and sweep the population Sharpe ratio across $SR \in \{0.5, 0.7, 1.0, 1.2, 1.5, 1.8, 2.0\}$. The DGP is IID Normal and three candidate models are compared: IID Normal, IID general and AR(1) Normal. Since the DGP is the simplest possible case, all three models are formally correctly specified (the IID general and AR(1) Normal models nest the IID Normal one). The aim is therefore not to study misspecification, which is absent by construction, but to show how the level of the Sharpe ratio interacts with the noise injected by the additional parameters estimated under the richer models. Figure 4c reports the resulting coverage as a function of SR . The iid normal model stays inside the admissible bands for all the sr values, while the iid non-normal shows a clear positive relation with the levels on sr, the ar normal model undercovers for all the sr values. The reason is the small sample size $T = 100$, as SR grows the richer models get noisier estimates of their parameters (like ϕ).

3.2.4 Scenario a4: non-existence of the fourth moment

When the DGP is IID with Student- t innovations and $\nu < 4$ degrees of freedom, the fourth moment does not exist and the population kurtosis is infinite. This violates one of the regularity conditions of the Method of Moments used to derive the asymptotic variance of the Sharpe ratio estimator: for $V_{\widehat{SR}}$ to be finite, we need $\mathbb{E}[r_t^4] < \infty$. When this assumption fails, the asymptotic variance formula loses its theoretical justification and any confidence interval built from it inherits the same problem.

Figure 4d shows the experiment with an IID Student- t DGP and the correctly-specified IID Student- t candidate model, sweeping the degrees of freedom across $\nu \in \{50, 6, 4, 3.5, 3\}$. As shown in Section A.2, the tail index of the IID Student- t coincides with ν , so the configuration $\nu = 4$ sits exactly at the boundary where the fourth moment ceases to exist. For $\nu = 50$ and $\nu = 6$, the coverage curves converge to the nominal level as standard asymptotic theory predicts. For $\nu < 4$, however, the qualitative behavior fundamentally shifts: at $\nu = 3.5$, convergence is really slow, and at $\nu = 3$, the coverage drifts *away* from the nominal level as the sample size T grows. This empirical behavior aligns with the theoretical results of Kao et al., 2025 (Lemma 2.3), who demonstrate that for a tail index $2 < \kappa < 4$, the estimated Sharpe ratio converges at a rate of T^ϕ , where $\phi = 1 - 2/\kappa$. Because $\kappa < 4$, the exponent ϕ is strictly less than $1/2$, resulting in a convergence rate that is mathematically slower than the standard $\sqrt{T}$ rate, they also prove that the limiting distribution incorporates a non-vanishing, κ -dependent bias term, which theoretically explains the observed divergence in coverage.

3.2.5 Scenario a5: efficiency

The last scenario inverts the logic. The DGP is IID Normal, the simplest possible case, and we compare candidate models of increasing complexity: IID Normal, IID Student- t , IID general, AR(1) Normal and AR(1) general. All five models nest the true DGP, so all five are correctly specified; what the experiment isolates is the cost of overparameterization. Each additional parameter that is not actually present in the DGP introduces estimation noise into the variance formula, which in finite samples produces wider, more volatile confidence intervals and a loss of efficiency.

Figure 4e reports the resulting coverage as a function of T . At the smallest sample size ($T = 100$), the more complex models do not merely lose precision: some of them fail to attain the nominal coverage at all, because the estimation of nuisance parameters that are zero by construction (autocorrelation, skewness, excess kurtosis) introduces a finite-sample error that dominates any

theoretical gain from flexibility. As T grows the gap closes and all curves approach the nominal level, but the practical lesson is clear: in small samples, the leanest variance formula compatible with the data-generating mechanism is the preferable one, and the price of insurance against possible structures that are absent from the data can be substantial.

3.3 HAC Estimator

The parametric models considered so far require assumptions about the distribution and the dependence structure of the returns. A non-parametric alternative is the HAC (Heteroskedasticity and Autocorrelation Consistent) estimator, introduced by Newey and West, 1987 and refined by Andrews, 1991, which estimates the asymptotic variance without imposing any assumption on the functional form of the heteroscedasticity or on the order of the autocorrelation.

Let $m_t = m(r_t, \theta)$ be the vector of moment functions evaluated at observation t , as defined in (1). The long-run variance matrix is

$$S = \sum_{j=-\infty}^{\infty} \text{Cov}(m_t, m_{t-j}) = \Gamma_0 + \sum_{j=1}^{\infty} (\Gamma_j + \Gamma_j^\top), \quad (23)$$

where $\Gamma_j = \text{Cov}(m_t, m_{t-j})$. The HAC estimator replaces this infinite sum by a truncated and weighted version,

$$\hat{\Sigma}_{HAC} = \hat{\Gamma}_0 + \sum_{j=1}^{b_T} k\left(\frac{j}{b_T}\right) (\hat{\Gamma}_j + \hat{\Gamma}_j^\top), \quad (24)$$

where $k(\cdot)$ is a kernel function that assigns decreasing weights to higher lags, b_T is the bandwidth, which grows with T , and $\hat{\Gamma}_j = \frac{1}{T} \sum_{t=j+1}^T \hat{m}_t \hat{m}_{t-j}^\top$ are the estimated sample autocovariances. The asymptotic variance of the Sharpe ratio is then obtained by applying the delta method to $\hat{\Sigma}_{HAC}$.

We use the Bartlett kernel (proposed by Newey and West, 1987), which guarantees that the estimated variance matrix is positive semidefinite in finite samples. Its weighting function decays linearly,

$$k(x) = \begin{cases} 1 - |x|, & \text{if } |x| \leq 1 \\ 0, & \text{otherwise,} \end{cases}$$

so that the weight applied to lag j is exactly $1 - (j/b_T)$.

The choice of the bandwidth b_T is decisive for the performance of the estimator. To avoid arbitrary choices, we use the automatic selection method of Andrews, 1991, which sets the number of lags by minimizing the Asymptotic Mean Squared Error (AMSE). For the Bartlett kernel, the resulting data-driven bandwidth is

$$b_T = 1.1447 (\alpha(1)T)^{1/3},$$

where $\alpha(1)$ is a population quantity that measures the strength of the autocorrelation present in the underlying moment conditions. The more persistent the data, the larger $\alpha(1)$ becomes, which calls for a larger number of lags b_T to capture the full temporal dependence. We estimate $\alpha(1)$ following Section 6 of Andrews, 1991, specifically equations (6.2) and (6.4).

The main appeal of the HAC estimator is that it requires neither a parametric model for the temporal dependence nor one for the distribution of the returns, which makes it robust to misspecification. The price, as we show next, is a loss of efficiency: when a parametric model is correctly specified its variance is the smallest attainable, and HAC pays for its generality by giving up part of that precision.

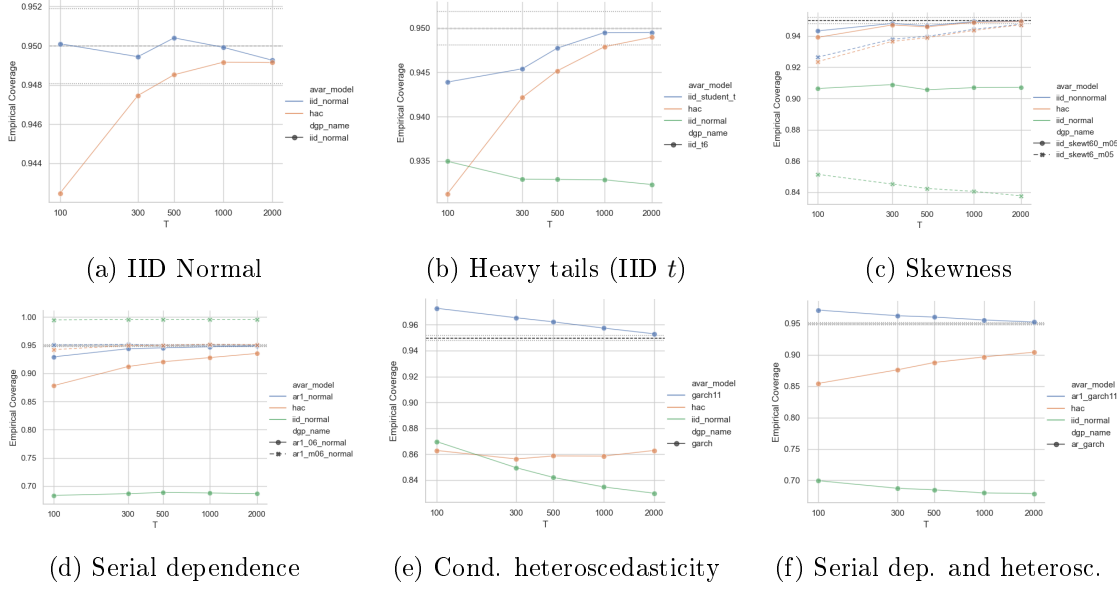


Figure 5: Empirical coverage of the nominal 95% confidence interval for the Sharpe ratio estimator as a function of the sample size T , comparing three variance estimators across the six DGPs of the stylized-fact scenarios. In each panel the blue curve is the correctly-specified parametric model, the orange curve is the HAC estimator, and the green curve is the misspecified IID Normal formula (which coincides with the correct model only in panel (a)). The horizontal band around 0.95 is the binomial acceptance region of (53), and all curves are based on $n = 50,000$ Monte Carlo replications.

Figure 5 compares the empirical coverage of the three approaches across the same DGPs used in the stylized-fact scenarios. The general pattern confirms the efficiency argument: the correctly-specified model converges to the nominal level faster than HAC in every case, while the misspecified IID Normal formula converges more slowly than HAC, the only exceptions appearing at $T = 100$, where the green curve occasionally sits slightly closer to the target than HAC. The HAC estimator therefore occupies an intermediate position. It is consistent, so it converges to the nominal coverage as the sample grows, but it forfeits the efficiency that the correct parametric model enjoys, which is the expected cost of dispensing with a parametric specification.

The exception to this ordering appears in the GARCH and AR-GARCH panels (Figures 5e and 5f), where HAC fails to reach satisfactory coverage even at the largest sample sizes considered. An internal inspection of the bandwidth selection is informative: in the GARCH case the number of lags chosen by the Andrews, 1991 procedure ranges only from 2 to 6, whereas in the AR case it ranges from 6 to 11. The markedly shorter bandwidths selected under GARCH suggest that the automatic procedure is not well suited to this setting. The same degradation of HAC inference under GARCH-type dependence with the Andrews, 1991 bandwidth has been documented by Ledoit and Wolf, 2008.

A caveat applies to these last two panels. For the GARCH and AR-GARCH DGPs the parametric models are evaluated with theoretical moments, following the convention explained in Section 3.2. This choice does not affect the HAC curve, which is non-parametric by construction and never uses θ_{DGP} , but it does place the blue parametric curve under an idealized condition. If the GARCH parameters would have been estimated from each sample, the noise in $\hat{\theta}$ would have degraded the parametric coverage as well, so the gap between the correctly-specified model and HAC displayed in these two panels overstates the advantage that the parametric approach would enjoy in practice.

4 Finite-Sample Bias

In the previous sections we derived the asymptotic distribution of the sample Sharpe ratio under different assumptions about the return-generating process. The sample mean and variance estimators are consistent, $\hat{\mu} \xrightarrow{P} \mu$ and $\hat{\sigma}^2 \xrightarrow{P} \sigma^2$, if we define $g(m, v) = m/\sqrt{v}$, which is continuous almost everywhere (except at $\sigma^2 = 0$). By the continuous mapping theorem (Proposition 2.27 of White, 2001), $\widehat{SR} = g(\hat{\mu}, \hat{\sigma}^2) \xrightarrow{P} g(\mu, \sigma^2) = SR$, so $\widehat{SR}$ is consistent. In practice, however, we work with finite samples, and there the estimator $\widehat{SR}$ has a bias that can matter when T is small. This section studies the source of that bias and derives analytical corrections.

4.1 Christie's (2005) bias approximation

The sample Sharpe ratio estimator is

$$\widehat{SR} = \frac{\hat{\mu}}{\hat{\sigma}},$$

where $\hat{\mu}$ is the sample mean and $\hat{\sigma}$ the sample standard deviation. Under normality both estimators are independent (see Theorem 5.3.1 of Casella and Berger, 2002), so that

$$E[\widehat{SR}] = \mu \cdot E\left[\frac{1}{\hat{\sigma}}\right].$$

The bias is fully determined by $E[1/\hat{\sigma}]$, since $\hat{\mu}$ is unbiased for μ .

Remark on the denominator. Christie uses the estimator with denominator T , that is, $\hat{\sigma}_T^2 = T^{-1} \sum_t (r_t - \bar{r})^2$, following Sharpe, 1994 (see his footnote 10). The detail matters because $\hat{\sigma}_T$ and $\hat{\sigma}_{T-1}$ have different first-order biases (see Appendix A.3).

Christie, 2005 proposes expanding $f(s) = 1/s$ around σ :

$$f'(s) = -\frac{1}{s^2}, \quad f''(s) = \frac{2}{s^3}.$$

The second-order Taylor expansion evaluated at σ gives

$$\frac{1}{\hat{\sigma}} \approx \frac{1}{\sigma} - \frac{1}{\sigma^2}(\hat{\sigma} - \sigma) + \frac{1}{\sigma^3}(\hat{\sigma} - \sigma)^2,$$

and taking expectations:

$$E\left[\frac{1}{\hat{\sigma}}\right] \approx \frac{1}{\sigma} - \frac{1}{\sigma^2}(E[\hat{\sigma}] - \sigma) + \frac{1}{\sigma^3}\text{Var}(\hat{\sigma}). \quad (25)$$

In his Appendix C, Christie claims that $E[\hat{\sigma}] = \sigma$, that is, that $\hat{\sigma}$ is unbiased. This is *incorrect*: by Jensen's inequality, since the square root is concave,

$$E[\hat{\sigma}] = E\left[\sqrt{\hat{\sigma}^2}\right] < \sqrt{E[\hat{\sigma}^2]} \leq \sigma,$$

so $\hat{\sigma}$ underestimates σ on average, regardless of which of the two estimators ($\hat{\sigma}_T$ or $\hat{\sigma}_{T-1}$) is used. Quantitatively, Appendix A.3 shows that

$$E[\hat{\sigma}_{T-1}] - \sigma \approx -\frac{\sigma}{4T}, \quad E[\hat{\sigma}_T] - \sigma \approx -\frac{3\sigma}{4T},$$

both biases of order $O(T^{-1})$.

Under the wrong unbiasedness assumption, the second term of (25) vanishes. For the third term, Appendix A.3 proves that $\text{Var}(\hat{\sigma}) \approx \sigma^2/(2T)$ to first order, the same for $\hat{\sigma}_T$ and $\hat{\sigma}_{T-1}$. Substituting:

$$E\left[\frac{1}{\hat{\sigma}}\right] \approx \frac{1}{\sigma} \left(1 + \frac{1}{2T}\right), \quad (26)$$

and therefore:

$$E[\widehat{SR}] \approx SR \left(1 + \frac{1}{2T}\right). \quad (\text{Christie}) \quad (27)$$

4.2 Correct approximation

We now present the correct approximation of Jobson and Korkie, 1981, in which the bias of $\hat{\sigma}$ is not ignored. We write the ratio $\widehat{SR} = \hat{X}/\hat{Y}$ directly, with $\hat{X} = \hat{\mu}$, $X = \mu$, $\hat{Y} = \hat{\sigma}$ and $Y = \sigma$, and define $\delta X = \hat{X} - X$ and $\delta Y^r = (\hat{Y} - Y)^r$:

$$\frac{\hat{X}}{\hat{Y}} = \frac{X \left(1 + \frac{\delta X}{X}\right)}{Y \left(1 + \frac{\delta Y}{Y}\right)} = \frac{X}{Y} \left(1 + \frac{\delta X}{X}\right) \left(1 + \frac{\delta Y}{Y}\right)^{-1}.$$

Writing it as a geometric series:

$$\left(1 + \frac{\delta Y}{Y}\right)^{-1} = 1 - \frac{\delta Y}{Y} + \frac{\delta Y^2}{Y^2} - \frac{\delta Y^3}{Y^3} + \frac{\delta Y^4}{Y^4} - \dots$$

Multiplying both factors:

$$\begin{aligned} \frac{\hat{X}}{\hat{Y}} = \frac{X}{Y} & \left(1 - \frac{\delta Y}{Y} + \frac{\delta Y^2}{Y^2} - \frac{\delta Y^3}{Y^3} + \frac{\delta Y^4}{Y^4} \right. \\ & \left. + \frac{\delta X}{X} - \frac{\delta X \delta Y}{XY} + \frac{\delta X \delta Y^2}{XY^2} - \frac{\delta X \delta Y^3}{XY^3} + \dots \right). \end{aligned}$$

Taking expectations uses two facts: (i) $E[\delta X] = 0$ because $\hat{\mu}$ is unbiased; (ii) $E[\delta X \delta Y^k] = E[\delta X] \cdot E[\delta Y^k] = 0$ for every $k \geq 1$, by independence between $\hat{\mu}$ and $\hat{\sigma}$ under normality. All cross terms drop out and we obtain:

$$E\left[\frac{\hat{X}}{\hat{Y}}\right] = \frac{X}{Y} \left(\underbrace{1 - \frac{E[\delta Y]}{Y} + \frac{E[\delta Y^2]}{Y^2}}_{\text{bias of order } O(T^{-1})} - \underbrace{\frac{E[\delta Y^3]}{Y^3} + \frac{E[\delta Y^4]}{Y^4} + \dots}_{\text{bias of order } O(T^{-2})} \right). \quad (28)$$

The orders are justified as follows: the bias $E[\hat{\sigma}] - \sigma = O(T^{-1})$ and the variance $\text{Var}(\hat{\sigma}) = O(T^{-1})$ add up to a net correction of order $O(T^{-1})$. In the same way, the second pair is $O(T^{-2})$, and so on.

Keeping only the $O(T^{-1})$ terms, and using the values computed in Appendix A.3,

$$-\frac{E[\delta Y]}{Y} \approx \begin{cases} \frac{1}{4T} & \text{with } \hat{\sigma}_{T-1}, \\ \frac{3}{4T} & \text{with } \hat{\sigma}_T, \end{cases} \quad \frac{E[\delta Y^2]}{Y^2} \approx \frac{1}{2T} \text{ in both cases,}$$

we obtain the correct result:

$$E[\widehat{SR}] \approx \begin{cases} SR \left(1 + \frac{3}{4T}\right) & \text{with denominator } T-1, \\ SR \left(1 + \frac{5}{4T}\right) & \text{with denominator } T. \end{cases} \quad (29)$$

The first one matches Jobson and Korkie, 1981 and Bao and Ullah, 2006.

Comparison with Christie. Christie reports the factor $1 + 1/(2T)$ (27). But the correct answer is:

- With denominator $T - 1$, the correct factor is $1 + 3/(4T)$: Christie underestimates the bias by a factor of $3/2$.

- With denominator T (the one he actually uses) the correct factor is $1 + 5/(4T)$: the under-estimation is by a factor of $5/2$.

In both cases Christie's correction is *smaller in magnitude* than the estimator's true bias.

For completeness we add the result of Miller and Gehr, 1978, which gives the exact bias (for $\hat{\sigma}_{T-1}$):

$$E[\widehat{SR}] = SR \cdot \sqrt{\frac{T-1}{2}} \frac{\Gamma(\frac{T-2}{2})}{\Gamma(\frac{T-1}{2})}. \quad (30)$$

4.3 Empirical comparison of the corrections

Figure 6 shows an experiment in which 500 000 IID Normal paths were simulated, comparing four estimators: no correction (black line, the simple point estimate), Christie's correction (blue line), the correct correction (green line), and the exact correction (orange line).

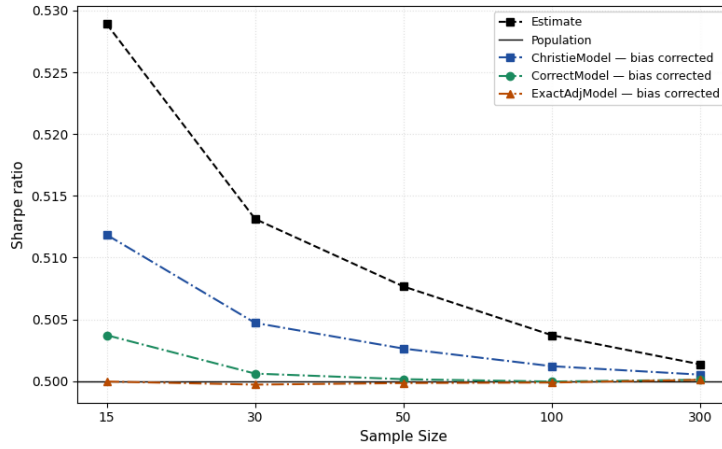


Figure 6: Empirical comparison of the bias corrections: no correction, Christie's correction (27), correct correction (29), and exact correction (30).

We see empirically what we proved analytically: Christie underestimates the true bias. Two points are worth stressing. First, both Christie's correction and the correct first-order correction keep only the $O(T^{-1})$ terms and drop the $O(T^{-2})$ contributions, while the orange line is the exact solution (30). Second, the gap between the correct first-order correction (green) and the exact result (orange) is very small, which suggests that the higher-order terms are not very important in this setting. As T grows, all estimates converge, since $\widehat{SR}$ is a consistent (asymptotically unbiased) estimator with or without correction.

4.4 General bias through the long-run variance

We can obtain a general expression for the bias of $\widehat{SR}$ using the results of Section 2.2 (the long-run variance S).

$$E[\widehat{SR}] - SR \approx \frac{SR}{2(T-1)} \left(\frac{S_{11}}{\sigma^2} - 1 \right) + \frac{1}{2T} \left(-\frac{S_{12}}{\sigma^3} + \frac{3SR}{4\sigma^4} S_{22} \right). \quad (31)$$

Proof. See Appendix A.4

This expression is fully general and depends only on the elements of S , which in turn depend on the assumed model for the returns. Solving for SR (which we will call $\widehat{SR}_{\text{corrected}}$):

$$\widehat{SR}_{\text{corrected}} = \left(\widehat{SR} + \frac{1}{2T} \frac{S_{12}}{\sigma^3} \right) \left[1 + \frac{1}{2(T-1)} \left(\frac{S_{11}}{\sigma^2} - 1 \right) + \frac{3}{8T\sigma^4} S_{22} \right]^{-1} \quad (32)$$

4.5 Bias corrections by model

We now apply formula (32) to the specific models of Section 2.2, substituting the elements S_{11} , S_{12} and S_{22} of the AR(1)-GARCH(1,1) model and obtaining the special cases ($\phi = 0$ or $\alpha = \beta = 0$) by direct restriction.

Table A.3 summarizes the bias-correction formulas for each model. The third column shows the corrected estimator found in (32).

Figure 7 contains one experiment per model, comparing the convergence of the unadjusted and the adjusted estimators; in every case there is an improvement (except for the GARCH, where the correction does not work for small T but then appears to introduce convexity and converges faster).

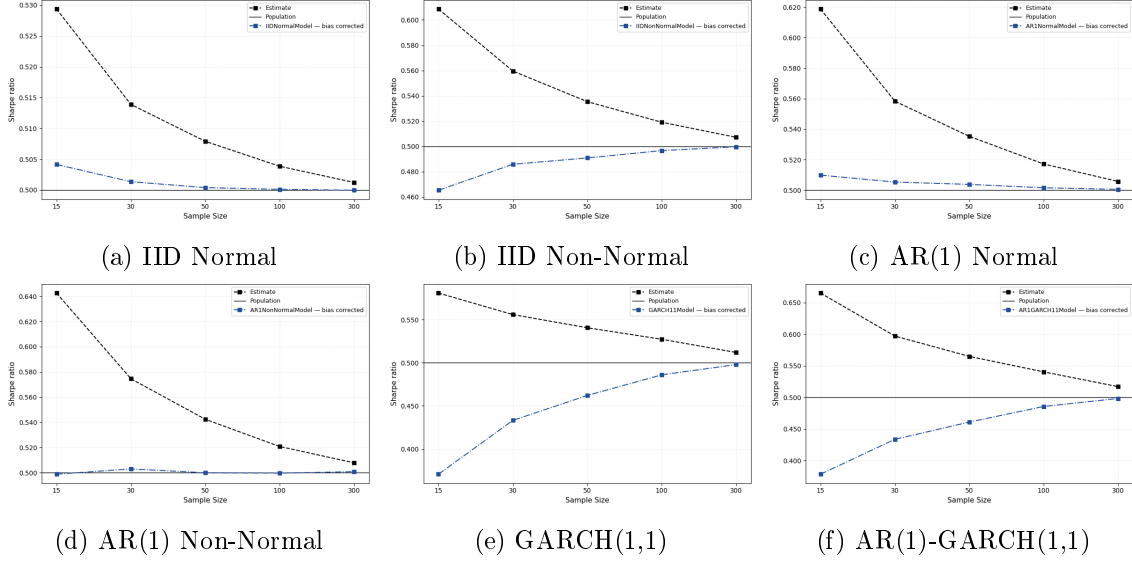


Figure 7: Individual comparison of the improvement from the bias correction.

5 Empirical application

In the previous chapters we analysed the Sharpe ratio estimator $\widehat{SR}$ on synthetic environments, where the true parameters and moments of the data-generating process (DGP) were known by construction. The aim of the present section is to apply the framework to real financial data. For each return series a parsimonious model is selected through a transparent diagnostic procedure; the resulting parameter estimates are then plugged into the asymptotic variance (AVAR) formulas derived in Section 2.2, and we analyze how the choice of model affects the inference on the Sharpe ratio.

5.1 Data

The dataset consists of weekly log returns over the most recent ten years, sourced from Reuters, for three SPDR exchange-traded funds:

- **XLV:** Health Care Select Sector SPDR.
- **XLRE:** Real Estate Select Sector SPDR.
- **SPIP:** SPDR Portfolio TIPS ETF (Treasury inflation-protected securities).

These three series were chosen because each exhibits both serial correlation in the level and conditional heteroskedasticity in the squared residuals, the two features that the AR(1)-GARCH(1,1) model is designed to capture. They therefore offer a setting in which the AVAR derived under iid Normal returns Lo, 2002, AR(1) returns López de Prado et al., 2025, and GARCH(1,1) returns López de Prado et al., 2026 can be contrasted on data where the simpler specifications are demonstrably insufficient. The empirical question is whether plugging in an AVAR formula that ignores one or both of these dynamic features leads to a noticeable over- or under-estimation of the sampling precision of $\widehat{SR}$. The risk-free rate is set to zero throughout.

5.2 Model specification

Procedure. For each series $\{r_t\}_{t=1}^T$, this three-step protocol is applied: Step 1 tests for serial dependence in the level via the Ljung-Box statistic Ljung and Box, 1978; if the test rejects, the conditional mean includes an AR(1) term, and is constant otherwise. Step 2 fits the mean equation alone and tests for ARCH effects in the residuals via the Lagrange Multiplier test of Engle, 1982; if the test rejects, the conditional variance is modelled as GARCH(1,1), and as a constant otherwise. Step 3 fits the full conditional model under Gaussian innovations and inspects the standardised residuals: if a Jarque-Bera test fails to reject normality, the innovation distribution is Normal; otherwise, if a separate symmetry test on the standardised residuals fails to reject the null of zero skewness, the distribution is Student- t ; otherwise it is skew- t . The protocol is summarised in Figure A.1.

Hierarchical testing versus information criteria. A natural alternative would be to fit a grid of candidate models (combinations of constant or AR(1) mean, constant or GARCH(1,1) variance, Normal, Student- t or skew- t innovations) and select by AIC or BIC. This route is deliberately avoided. When the candidate specifications are strictly nested, any information-criterion comparison in the grid is equivalent to a likelihood-ratio test in which the penalty difference plays the role of a critical value (Sin and White, 1996; Vuong, 1989). The order of magnitude of the penalty required for consistent selection is determined by theory, but the exact penalty, and therefore the implicit size of the test, is not (Sin and White, 1996, p. 224). A hierarchical protocol based on explicit specification tests (Ljung-Box for serial dependence, Engle’s LM test for ARCH effects) makes this choice transparent: the significance level is fixed ex ante, each test has known size and power against the alternative it targets, and the inclusion of each component is justified by direct evidence in the dimension it concerns.

Results. Table 3 reports the outcome of the three steps for the three series.

Table 3: Hierarchical specification, three SPDR series. P-values are reported as asterisks ($*p < 0.10$, $**p < 0.05$, $***p < 0.01$); $\hat{g}_1$ and $\hat{g}_2$ are the sample skewness and excess kurtosis of the Gaussian-model standardised residuals.

Series	Step 1		Step 2		Step 3			
	LB(r)	AR?	LM _{ARCH}	GARCH?	JB	skew	$\hat{g}_1$	$\hat{g}_2$
XLV	**	yes	***	yes	***	***	−0.446	1.509
XLRE	***	yes	***	yes	***	***	−0.489	1.496
SPIP	***	yes	***	yes	***	***	−0.538	2.406

For all three series the procedure selects the same final specification: an AR(1) conditional mean, a GARCH(1,1) conditional variance, and skew- t innovations.

5.3 Inference under alternative DGPs

Given a DGP $\mathcal{M}$ with parameter vector θ , the asymptotic variance $V_{\mathcal{M}}(\theta)$ of the Sharpe ratio estimator yields the probabilistic Sharpe ratio³, defined as the probability that the true Sharpe ratio exceeds a benchmark value SR^* :

$$\widehat{\text{PSR}}(\text{SR}^*) = \Phi\left(\frac{\widehat{\text{SR}} - \text{SR}^*}{\sqrt{V_{\mathcal{M}}(\hat{\theta})/T}}\right), \quad (33)$$

where $\widehat{\text{SR}}$ denotes the sample Sharpe ratio and Φ the standard normal cumulative distribution function. The benchmark is set at $\text{SR}^* = 0$, so that the PSR reduces to one minus the one-sided p-value for the null that the true Sharpe ratio is zero, computed under the asymptotic distribution implied by $\mathcal{M}$. The estimator $\widehat{\text{SR}}$ is invariant across DGPs; what changes across rows of Table 4 is only the AVAR plugged into (33).

Comparison of model specifications. Four specifications are compared:

1. iid Normal.
2. AR(1).
3. GARCH(1,1).
4. AR(1)-GARCH(1,1).

The progression (1)→(4) introduces one feature at a time and is intended to isolate the contribution of the AR-GARCH dynamics, which constitutes the novelty relative to the previous literature.

Results. Table 4 reports $\widehat{\text{PSR}}$ in percent for each (series, DGP) pair under $\text{SR}^* = 0$. The AVAR formulas are taken from Section 2 and the parameters are estimated through the procedure of Section ??.

Three observations stand out. First, the differences across DGPs are modest in magnitude: XLV is significant at the 5% level under all four specifications, while XLRE and SPIP are not significant under any. Second, the impact of the DGP is non-monotonic in model complexity. Relative to iid Normal, the introduction of an AR component alone (column 2) raises the PSR for XLV and XLRE but lowers it for SPIP, while the introduction of GARCH alone (column 3) lowers the PSR for all three series. The richer AR(1)-GARCH(1,1) specification lands between these two corrections. The

³The probabilistic Sharpe ratio was originally proposed by Bailey and Lopez de Prado, 2012.

Table 4: $\widehat{\text{PSR}}$ (%) for $\text{SR}^* = 0$ under four alternative DGPs.

Series	iid Normal	AR(1)	GARCH(1,1)	AR(1)-GARCH(1,1)
XLV	95.80	96.95	94.94	96.22
XLRE	76.51	79.89	76.14	78.72
SPIP	42.88	42.20	42.80	42.21

implication is that more flexible DGPs do not uniformly produce more conservative inference; the direction of the correction depends on the joint estimation of the parameters. Third, XLV sits closest to a decision boundary: its iid-Normal PSR is 95.80%, just past the conventional 95% threshold, and the AR(1)-GARCH(1,1) specification raises it to 96.22%. *Esperaba encontrar una serie en la que un DGP más simple produjera un Sharpe significativo, mientras que la especificación más compleja invirtiera la conclusión, esto pasa porque la magnitud de phi es pequeña (debería reportar los parámetros estimados?). los SR tambien son pequeños porque estoy con datos semanales, en los mensuales no habia autocorrelacion (imagino que relacionado con la teoria de eficiencia de mercado, aunque he visto que un autor que usa los indices HFR (que son sobre hedge funds) reporta que sí tienen autocorr y bastante grande).*

6 Alternative Metrics

The classical Sharpe ratio summarises performance through the first two moments of the return distribution. When returns exhibit pronounced skewness or heavy tails, features common in many real-world strategies (Cont, 2001), this ratio may misrepresent the underlying risk-return profile. Several alternative metrics have been proposed to incorporate higher-moment information; we begin with the most direct approach, in which the skewness is built explicitly into the metric itself.

6.1 Adjusted for Skewness Sharpe Ratio (ASSR)

Within Expected Utility Theory, this can be done by Taylor-expanding the certainty equivalent of the return distribution around its mean. Assuming an investor with constant absolute risk aversion (CARA), Zakamouline and Koekebakker, 2009 carry out this expansion and retain terms up to the third central moment, yielding the *Adjusted for Skewness Sharpe Ratio*

$$ASSR = SR \sqrt{1 + \frac{\gamma}{3} SR}, \quad (34)$$

where $SR = \mu/\sigma$ and γ are the population Sharpe ratio and the population skewness of returns. The radicand falls below one for $\gamma < 0$ and rises above one for $\gamma > 0$, so ASSR penalises (rewards) negatively (positively) skewed strategies. The expansion is well-defined while $1 + \gamma SR/3 > 0$ ⁴.

The aim of this section is to compare two complementary ways of accounting for skew: building it explicitly into the metric, as ASSR does, versus retaining the classical Sharpe ratio as the point estimate but performing statistical inference under a return model that admits skewness. We ask whether these two routes imply the same preference ordering over portfolios.

6.1.1 The Probabilistic Sharpe Ratio (PSR)

López de Prado et al., 2025 formulates the probabilistic sharpe ratio (PSR), defined as the probability that $\widehat{SR}$ is greater than a predefined benchmark SR^* . We take $SR^* = 0$ throughout. From the asymptotic normality of $\widehat{SR}$ derived in Section 2.1,

$$\sqrt{T}(\widehat{SR} - SR) \xrightarrow{d} \mathcal{N}(0, V_{\widehat{SR}}), \quad (35)$$

the PSR takes the form

$$PSR(SR^*) = \Phi\left(\frac{\widehat{SR} - SR^*}{\sqrt{V_{\widehat{SR}}/T}}\right). \quad (36)$$

The asymptotic variance $V_{\widehat{SR}}$ depends on the assumed data-generating process. We consider two specifications, both derived in Section 2.2:

PSR_{normal}. Assumes iid Gaussian returns, giving $V_{\widehat{SR}} = 1 + \frac{1}{2}SR^2$. Skewness does not enter the formula, so this estimator is blind to it.

PSR_{skew}. Assumes iid returns with skewness γ and excess kurtosis zero ($\kappa = 3$), giving $V_{\widehat{SR}} = 1 - \gamma SR + \frac{1}{2}SR^2$. A negative skew inflates the variance and lowers the PSR.

For the realisations of ASSR and the two PSR variants we plug $\widehat{SR}$ and $\widehat{\gamma}$ into SR and γ respectively, and throughout the rest of the section we simply write SR and γ . No estimation is performed here: we treat the point estimates $\widehat{SR}$ and $\widehat{\gamma}$ as given inputs rather than as quantities to be estimated.

Both ASSR and *PSR_{skew}* are monotone increasing in SR and decreasing in γ , but they weight the trade-off between the two differently. The natural question is whether the resulting rankings of any two portfolios always coincide. We answer it geometrically.

⁴Equivalently $\gamma > -3/SR$; assuming $SR > 0$, this means the metric is undefined for heavily negatively skewed returns.

6.1.2 Indifference curves

For a metric $M(SR, \gamma)$, the *indifference curve* at level c is the set of (SR, γ) pairs that the metric values equally,

$$\{(SR, \gamma) : M(SR, \gamma) = c\}.$$

Two portfolios on the same curve are considered equivalent under M ; one above (higher SR, or less negative skew) is preferred, and one below is dispreferred. Anchoring the level at a reference point (SR_{ref}, γ_{ref}) , the curve traces, for each SR, the value of skewness that the metric considers equivalent to that reference. A flatter curve indicates a more skew-averse metric: a larger SR increase is required to offset a given amount of additional negative skew.

For our three metrics with $SR^* = 0$, the indifference curves through (SR_{ref}, γ_{ref}) admit closed forms:

$$PSR_{normal} : \quad SR = SR_{ref} \quad (\text{vertical lines, skew is invisible}), \quad (37)$$

$$PSR_{skew} : \quad \gamma = \frac{SR_{ref}^2 - SR^2}{SR_{ref}^2 SR} + \gamma_{ref} \frac{SR}{SR_{ref}}, \quad (38)$$

$$ASSR : \quad \gamma = \frac{3(SR_{ref}^2 - SR^2)}{SR^3} + \gamma_{ref} \left(\frac{SR_{ref}}{SR} \right)^3. \quad (39)$$

Differentiating at $SR = SR_{ref}$ gives the local slopes

$$\left. \frac{d\gamma}{dSR} \right|_{PSR_{skew}} = -\frac{2 - \gamma_{ref} SR_{ref}}{SR_{ref}^2}, \quad \left. \frac{d\gamma}{dSR} \right|_{ASSR} = -\frac{6 + 3\gamma_{ref} SR_{ref}}{SR_{ref}^2}.$$

In the special case $\gamma_{ref} = 0$ these reduce to $-2/SR_{ref}^2$ and $-6/SR_{ref}^2$ respectively: locally, ASSR is exactly three times more permissive than PSR_{skew} . For $\gamma_{ref} < 0$ the ratio shrinks, but ASSR remains the more permissive of the two throughout the negatively-skewed region.

Figure 8 shows the curves through $(SR_{ref}, -1)$ for $SR_{ref} \in \{0.15, 0.25, 0.35, 0.45\}$. The grey band between the blue (PSR_{skew}) and red (ASSR) curves is the *disagreement zone*: a portfolio in that band is preferred to the reference under one metric and dispreferred under the other. Outside the band the two skew-aware metrics rank portfolios in the same order.

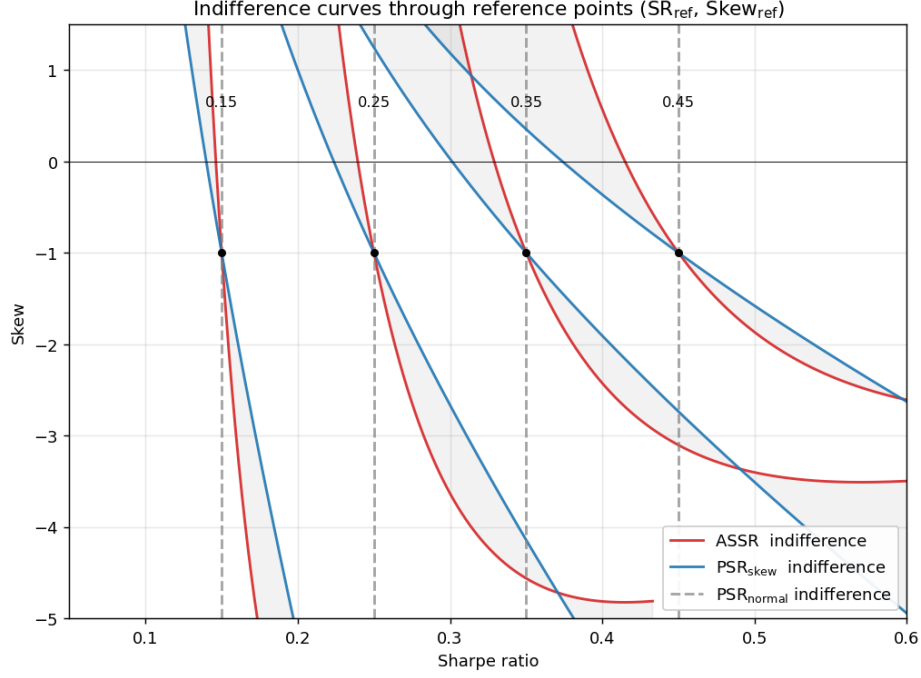


Figure 8: Indifference curves through reference points $(SR_{ref}, -1)$ for PSR_{normal} (grey dashed verticals), PSR_{skew} (blue), and ASSR (red). The grey band between the blue and red curves is the region in which the two skew-aware metrics rank a portfolio in opposite directions relative to the reference.

6.1.3 Numerical comparison

To make the implications concrete we construct eight portfolios with theoretical moments hand-picked to lie outside the grey disagreement zone of Figure 8, a configuration that is far from contrived and could plausibly arise in practice when comparing strategies with different skew profiles. The resulting metric values appear in Table 5 and the corresponding ranks in Table 6.

Portfolio	SR	γ	PSR_{normal}	PSR_{skew}	ASSR
P1	0.15	-1.00	0.8518	0.8351	0.1462
P2	0.16	-5.00	0.8671	0.7973	0.1370
P3	0.25	-1.00	0.9576	0.9390	0.2394
P4	0.30	-4.00	0.9800	0.9195	0.2324
P5	0.35	-1.00	0.9913	0.9804	0.3290
P6	0.40	-3.00	0.9965	0.9682	0.3098
P7	0.45	-1.00	0.9987	0.9943	0.4149
P8	0.50	-2.00	0.9995	0.9918	0.4082

Table 5: Theoretical moments and metric values for the eight test portfolios ($T = 50$, $SR^* = 0$).

Portfolio	$Rk\ PSR_{normal}$	$Rk\ PSR_{skew}$	$Rk\ ASSR$
P1	8	7	7
P2	7	8	8
P3	6	5	5
P4	5	6	6
P5	4	3	3
P6	3	4	4
P7	2	1	1
P8	1	2	2

Table 6: Rankings (1 = best) under each metric.

The Spearman correlations against the ASSR ranks make this explicit:

$$\begin{aligned}\rho(PSR_{skew}, ASSR) &= 1.0000, \\ \rho(PSR_{normal}, ASSR) &= 0.9048, \\ \Delta\rho &= +0.0952.\end{aligned}$$

PSR_{skew} captures exactly the asymmetric skew preference that ASSR was constructed to encode, while PSR_{normal} does not.

Conclusion. For portfolios outside the disagreement band of Figure 8, PSR_{skew} and ASSR yield identical rankings. A correctly specified version of the Sharpe-ratio inferential framework, one whose asymptotic variance accounts for skewness, therefore recovers the same ordering as the explicitly skew-adjusted estimator. The dedicated ASSR is not strictly necessary in this regime.

6.2 Joint distribution with the Value-at-Risk (VaR)

In Section 2.2 we derived the asymptotic variance of the Sharpe ratio estimator under several assumptions on the returns. Two practical considerations motivate complementing that study with a second risk metric. First, even if the correct model for the data is identified, the asymptotic variance of $\widehat{SR}$ depends on higher-order moments, such as the kurtosis, which are hard to estimate precisely (Cont, 2001). Second, the Sharpe ratio focuses on the potential gains per unit of risk (dispersion in general), while the left-tail risk under strong asymmetry is ignored.

Instead of contrasting the two metrics individually (as in the previous subsection), we will now use them in a complementary way and find their joint distribution. The methodology is identical to that of Section 2.1, the method of moments combined with the delta method (multivariate this time), though the inclusion of the quantile introduces a non-smooth moment condition that requires additional justification.

We define $VaR_\alpha = -q_\alpha > 0$ for $\alpha \in (0, 0.5)$ as the α -th quantile of the return distribution, that is, the value q_α such that $\mathbb{P}(r_t \leq q_\alpha) = \alpha$.

6.2.1 Non-parametric quantile estimator

We assume $r_t \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and add the quantile to the parameter vector, $\theta = (\mu, \sigma^2, q_\alpha)^\top$. The vector of moment conditions is $\mathbb{E}[m(r_t, \theta_0)] = 0$, where

$$m(r_t, \theta) = \begin{pmatrix} r_t - \mu \\ (r_t - \mu)^2 - \sigma^2 \\ \mathbb{1}\{r_t \leq q_\alpha\} - \alpha \end{pmatrix}, \quad (40)$$

The first two coincide with those used for the Sharpe ratio in Section 2.2.

Long-run variance. $S = \sum_{h=-\infty}^{\infty} \Gamma(h) = \text{Var}[m(r_t, \theta_0)]$ is now a 3×3 matrix whose upper 2×2 block coincides with the one computed for the SR.

The new elements are obtained using the change of variable $u = (r_t - \mu)/\sigma$, the identity $\phi'(u) = -u\phi(u)$, and the notation $z_\alpha = \Phi^{-1}(\alpha)$:

$$\begin{aligned} S_{13} &= \mathbb{E}[(r_t - \mu) \mathbb{1}\{r_t \leq q_\alpha\}] = \int_{-\infty}^{q_\alpha} (r_t - \mu) f(r_t) dr_t = \int_{-\infty}^{q_\alpha} (r_t - \mu) \frac{1}{\sigma} \phi\left(\frac{r_t - \mu}{\sigma}\right) dr_t \\ &= \int_{-\infty}^{z_\alpha} \sigma u \frac{1}{\sigma} \phi(u) \sigma du = \sigma \int_{-\infty}^{z_\alpha} u \phi(u) du = -\sigma \int_{-\infty}^{z_\alpha} \phi'(u) du = -\sigma [\phi(u)]_{-\infty}^{z_\alpha} \\ &= -\sigma \phi(z_\alpha). \end{aligned}$$

Integrating by parts and using $\Phi(z_\alpha) = \alpha$,

$$\begin{aligned} S_{23} &= \mathbb{E}[(r_t - \mu)^2 \mathbb{1}\{r_t \leq q_\alpha\}] - \sigma^2 \mathbb{E}[\mathbb{1}\{r_t \leq q_\alpha\}] \\ &= \int_{-\infty}^{q_\alpha} (r_t - \mu)^2 f(r_t) dr_t - \sigma^2 \alpha = \int_{-\infty}^{z_\alpha} \sigma^2 u^2 \frac{1}{\sigma} \phi(u) \sigma du - \sigma^2 \alpha \\ &= \sigma^2 \left[-u\phi(u) - \int -\phi(u) du \right]_{-\infty}^{z_\alpha} - \sigma^2 \alpha = \sigma^2 [\Phi(z_\alpha) - z_\alpha \phi(z_\alpha)] - \sigma^2 \alpha \\ &= -\sigma^2 z_\alpha \phi(z_\alpha). \end{aligned}$$

Finally,

$$\begin{aligned} S_{33} &= \mathbb{E}[(\mathbb{1}\{r_t \leq q_\alpha\} - \alpha)^2] = \mathbb{E}[\mathbb{1}\{r_t \leq q_\alpha\}] - 2\alpha \mathbb{E}[\mathbb{1}\{r_t \leq q_\alpha\}] + \alpha^2 \\ &= \alpha - 2\alpha^2 + \alpha^2 = \alpha(1 - \alpha). \end{aligned}$$

Therefore,

$$S = \begin{pmatrix} \sigma^2 & 0 & -\sigma \phi(z_\alpha) \\ 0 & 2\sigma^4 & -\sigma^2 z_\alpha \phi(z_\alpha) \\ -\sigma \phi(z_\alpha) & -\sigma^2 z_\alpha \phi(z_\alpha) & \alpha(1 - \alpha) \end{pmatrix}. \quad (41)$$

Expected Jacobian H .

$$H = \mathbb{E} \left[\frac{\partial m(r_t, \theta)}{\partial \theta^\top} \right]_{\theta=\theta_0},$$

is the *expectation of the derivative* of the moment condition. For the first two components of (40) this poses no problem: both are smooth in θ and contribute -1 on the diagonal, as in Section 2.2.

The third component contains the indicator $\mathbb{1}\{r_t \leq q_\alpha\}$, which as a function of q_α is a step function: its pointwise derivative is zero except at $r_t = q_\alpha$, so the formula $\mathbb{E}[\partial m_3 / \partial q_\alpha]$ taken literally would give zero and make the Jacobian singular. The relevant derivative is not that of the indicator but that of the already-integrated moment condition, which is smooth. Bahadur's representation of the empirical quantile (Bahadur, 1966) justifies that this is the derivative entering the asymptotic expansion:

$$\frac{\partial}{\partial q_\alpha} \mathbb{E}[\mathbb{1}\{r_t \leq q_\alpha\} - \alpha] = \frac{\partial}{\partial q_\alpha} [F(q_\alpha) - \alpha] = f(q_\alpha) = \frac{\phi(z_\alpha)}{\sigma}.$$

so,

$$H = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & \phi(z_\alpha)/\sigma \end{pmatrix}, \quad H^{-1} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & \sigma/\phi(z_\alpha) \end{pmatrix}. \quad (42)$$

Asymptotic covariance of θ . Unlike Section 2.2, here $H \neq -I_3$, so the sandwich $\Sigma_\theta = H^{-1}S(H^{-1})^\top$ is not trivial. Carrying out the products and using $f(q_\alpha) = \phi(z_\alpha)/\sigma$,

$$\Sigma_\theta = \begin{pmatrix} \sigma^2 & 0 & \sigma^2 \\ 0 & 2\sigma^4 & \sigma^3 z_\alpha \\ \sigma^2 & \sigma^3 z_\alpha & \frac{\sigma^2 \alpha(1-\alpha)}{\phi(z_\alpha)^2} \end{pmatrix}. \quad (43)$$

Delta method. The function of interest also changes and is $g : \mathbb{R}^3 \rightarrow \mathbb{R}^2$,

$$g(\theta) = \begin{pmatrix} SR \\ VaR \end{pmatrix} = \begin{pmatrix} \mu/\sqrt{\sigma^2} \\ -q_\alpha \end{pmatrix}, \quad G = \nabla_\theta g(\theta) = \begin{pmatrix} \frac{1}{\sigma} & -\frac{\mu}{2\sigma^3} & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Then $V_h^{np} = G \Sigma_\theta G^\top$ yields

$$\sqrt{T} \begin{pmatrix} \widehat{SR} - SR \\ \widehat{VaR} - VaR \end{pmatrix} \xrightarrow{d} \mathcal{N}(0, V_h^{np}), \quad V_h^{np} = \begin{pmatrix} 1 + \frac{SR^2}{2} & \frac{\mu z_\alpha}{2} - \sigma \\ \frac{\mu z_\alpha}{2} - \sigma & \frac{\sigma^2 \alpha(1-\alpha)}{\phi(z_\alpha)^2} \end{pmatrix}. \quad (44)$$

The superscript “ np ” distinguishes this estimator from its parametric counterpart, which we derive next.

6.2.2 Parametric Gaussian estimator

If we accept normality not only for inference but also for the *construction* of the VaR estimator itself, the location-scale equivariance of the quantile gives $q_\alpha = \mu + \sigma z_\alpha$, so that q_α is no longer a free parameter. The reduced parameter vector is $\theta = (\mu, \sigma^2)^\top$ and the moment conditions are exactly those used in the SR analysis alone: both are smooth and the indicator function disappears. All the results of Section 2.2 then apply up to the definition of the transformation, with $\Sigma_\theta = S = \text{diag}(\sigma^2, 2\sigma^4)$.

The transformation is now $h : \mathbb{R}^2 \rightarrow \mathbb{R}^2$,

$$h(\theta) = \begin{pmatrix} \mu/\sqrt{\sigma^2} \\ -(\mu + \sqrt{\sigma^2} z_\alpha) \end{pmatrix}, \quad G = \begin{pmatrix} \frac{1}{\sigma} & -\frac{\mu}{2\sigma^3} \\ -1 & -\frac{z_\alpha}{2\sigma} \end{pmatrix}.$$

A direct computation of $V_h^p = G \Sigma_\theta G^\top$ gives

$$V_{h,11}^p = 1 + \frac{SR^2}{2}, \quad V_{h,12}^p = \frac{\mu z_\alpha}{2} - \sigma, \quad V_{h,22}^p = \sigma^2 \left(1 + \frac{z_\alpha^2}{2} \right),$$

so that

$$V_h^p = \begin{pmatrix} 1 + \frac{SR^2}{2} & \frac{\mu z_\alpha}{2} - \sigma \\ \frac{\mu z_\alpha}{2} - \sigma & \sigma^2 \left(1 + \frac{z_\alpha^2}{2} \right) \end{pmatrix}. \quad (45)$$

6.2.3 Comparison and efficiency

Table 7 sets the two estimators side by side.

Two observations follow immediately. First, $V_{h,11}$ and $V_{h,12}$ are identical in the two estimators. This is structural: both quantities depend only on $(\hat{\mu}, \hat{\sigma}^2)$, which are estimated the same way in

Entry	Non-parametric V_h^{np}	Parametric V_h^p
(1, 1) $\text{AVar}(\widehat{SR})$	$1 + SR^2/2$	$1 + SR^2/2$
(1, 2) covariance	$\mu z_\alpha/2 - \sigma$	$\mu z_\alpha/2 - \sigma$
(2, 2) $\text{AVar}(\widehat{VaR})$	$\sigma^2 \alpha(1 - \alpha)/\phi(z_\alpha)^2$	$\sigma^2(1 + z_\alpha^2/2)$

Table 7: Asymptotic covariance entries for the VaR estimators via the empirical quantile (non-parametric) and the Gaussian form (parametric) when $r_t \sim \mathcal{N}(\mu, \sigma^2)$.

both approaches; the third moment condition of the non-parametric case adds information about q_α , but, by the block structure of Σ_θ , that information does not feed back into the variance of $\widehat{\mu}$ or that of $\widehat{\sigma}^2$. Second, the covariance entry $\mu z_\alpha/2 - \sigma$ is negative whenever $\mu z_\alpha < 2\sigma$, which, since $z_\alpha < 0$ for $\alpha < 0.5$, holds for any positive mean return and, in fact, also for moderately negative mean returns. The negative covariance reads as follows: shocks that raise the estimated mean push $\widehat{SR}$ up while shifting the left tail to the right, lowering $\widehat{VaR}$; that is, an overestimation of $\widehat{SR}$ comes together with an underestimation of $\widehat{VaR}$.

The only entry that differs is $V_{h,22}$; we define the asymptotic relative efficiency as

$$\frac{V_{h,22}^{np}}{V_{h,22}^p} = \frac{\alpha(1 - \alpha)}{\phi(z_\alpha)^2 (1 + z_\alpha^2/2)} > 1 \quad \forall \alpha \in (0, 0.5), \quad (46)$$

and it grows quickly as $\alpha \rightarrow 0$:

α	0.05	0.01	0.001
Efficiency ratio	≈ 1.90	≈ 3.77	≈ 11

Table 8: Asymptotic relative efficiency of the parametric VaR estimator against the empirical one at standard confidence levels under the (true) Normal model.

The intuition is direct. The empirical quantile uses only the small fraction of observations near the tail, and the density $f(q_\alpha)$ in the denominator measures the mass of data at that point, whereas the parametric estimator pools *all* the observations through $\widehat{\mu}$ and $\widehat{\sigma}$ and then “reads off” the quantile from the assumed Gaussian form. The deeper the tail, the scarcer the data and the wider the gap.

The bias–variance trade-off. Expression (46) is computed under the assumption that returns are truly Gaussian. If they are not, and financial returns rarely are, the parametric estimator becomes inconsistent: $\widehat{VaR}^p \xrightarrow{p} -(\mu + \sigma z_\alpha) \neq -q_\alpha^{\text{true}}$. No variance reduction can offset a first-order bias. The non-parametric estimator, by contrast, remains consistent for the true quantile of the return distribution.

Monte Carlo illustration. Figures 9 and 10 report a Monte Carlo experiment of 50 000 replications that visually contrasts the two estimators under correct specification (Gaussian returns) and under misspecification (Student- t_6 returns). Each panel shows the marginal distributions of $\widehat{SR}$ and $\widehat{VaR}$ on the axes and the two 95% asymptotic confidence ellipses overlaid at the center. Under Gaussian returns, both ellipses are centered at the true (SR, VaR) and the parametric ellipse is strictly contained in the non-parametric one, exactly as Table 8 predicts. The empirical coverage is 95.0% for the parametric ellipse and 95.4% for the non-parametric one, confirming that the parametric estimator reaches the nominal coverage faster (it is more efficient). Under Student- t_6 returns, by contrast, the parametric ellipse no longer covers the true value: its center shifts by about -0.044 (the asymptotic bias) and its empirical coverage collapses to 14.8%, while the non-parametric ellipse stays correctly centered with an empirical coverage of 95.4%.

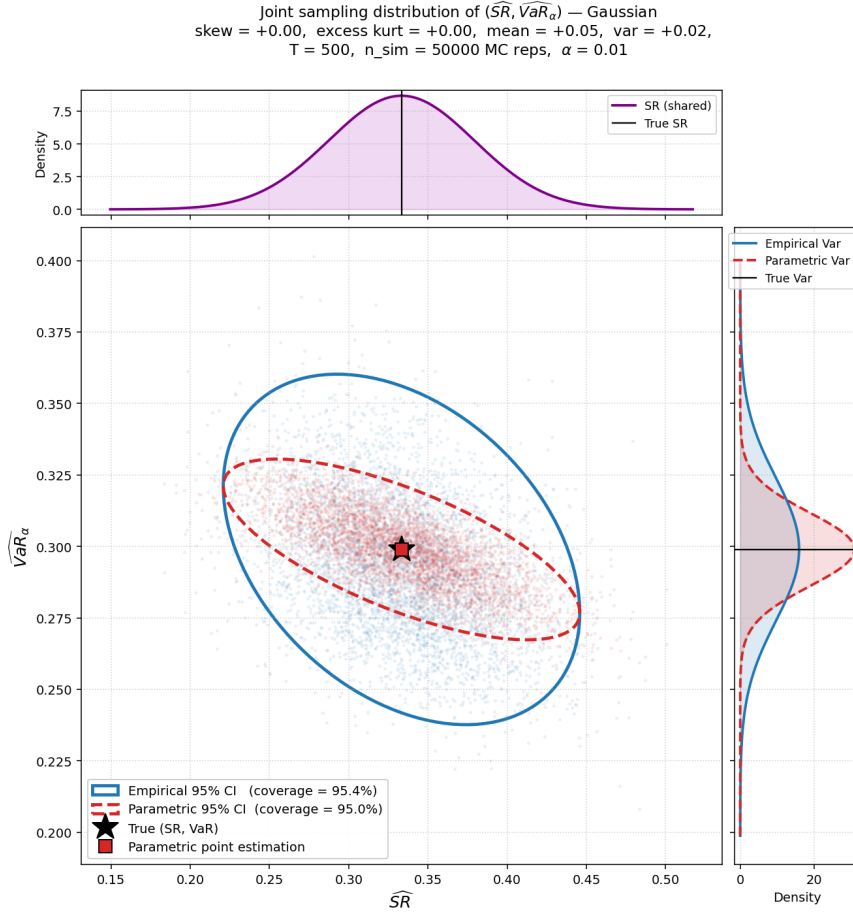


Figure 9: Joint distribution of $(\widehat{SR}, \widehat{VaR}_\alpha)$ under iid $\mathcal{N}(\mu, \sigma^2)$ returns. The parametric ellipse is contained in the non-parametric one, reflecting the efficiency gain of the Gaussian estimator under correct specification. Empirical 95% coverage: 95.0% (parametric) against 95.4% (non-parametric); 50 000 replications.

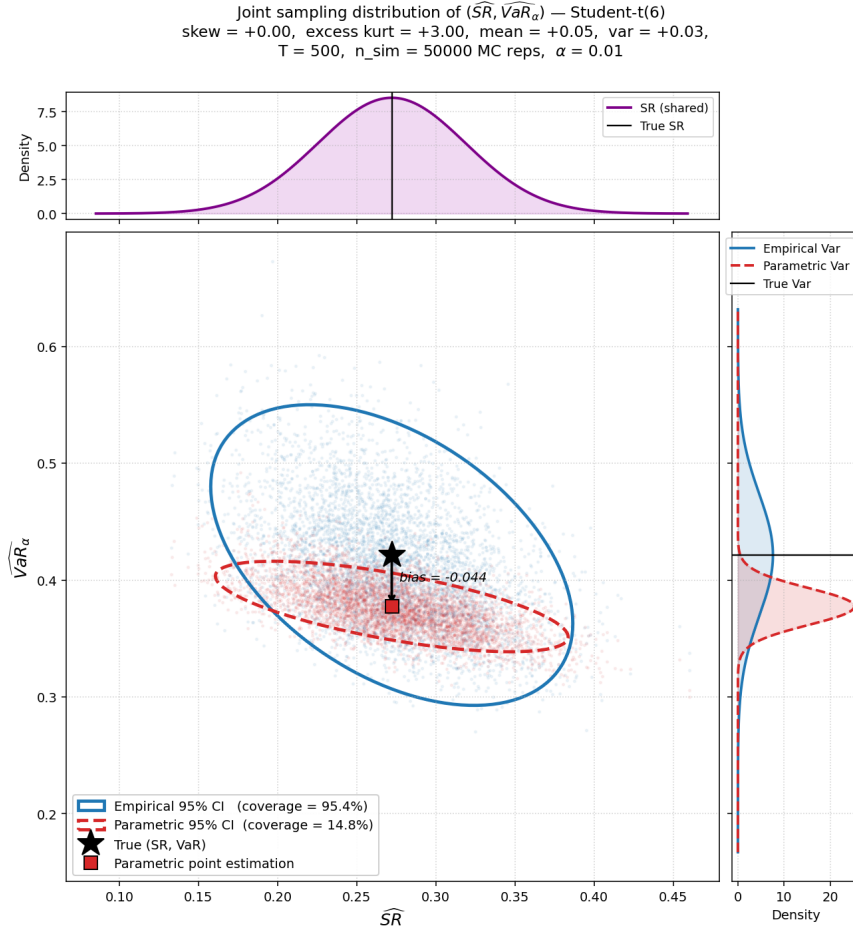


Figure 10: Joint distribution of $(\widehat{SR}, \widehat{VaR}_\alpha)$ under iid Student- t_6 returns (rescaled to match σ). The parametric ellipse is biased, shifted by -0.044 from the true value, and its empirical coverage collapses to 14.8%, while the non-parametric ellipse stays correctly centered with a coverage of 95.4%; 50 000 replications.

7 Conclusion

This work has studied the statistical behaviour of the Sharpe ratio estimator $\widehat{SR}$ along the three lines set out in the Introduction: the asymptotic variance, the coverage of the asymptotic confidence intervals, and the finite-sample bias, all under various configurations. It has also added two extensions that connect the metric to a skewness-adjusted criterion and to the Value-at-Risk.

The first concerns the asymptotic variance. The main analytical result is a semi-closed-form expression for the $\widehat{SR}$ variance under AR(1)-GARCH(1,1) returns with general innovations, derived in Appendix B; the qualifier *semi* reflects that the odd moments of the conditional variance have no known closed form and are handled with a first-order Taylor expansion. The sensitivity analysis singled out the population Sharpe ratio as the dominant driver of the variance, with the kurtosis and the skewness as the next two, the ϕ parameter concentrated its effect at the boundaries, while the volatility parameters α and β matter mostly near the integration frontier and have a moderate average effect across the admissible region.

The second examined, whether the intervals built from these formulas reach their nominal level. Misspecified intervals (those whose variance formula ignores heavy tails, skewness, serial dependence, or conditional heteroscedasticity that is actually present in the data) are systematically too narrow and undercover their nominal level. The autoregressive coefficient is the one exception worth flagging: positive persistence degrades coverage sharply, while negative persistence of the same magnitude leaves the interval near or even slightly above the nominal level. The non-parametric HAC estimator occupies an intermediate position between the correctly specified model and the misspecified one, consistent but less efficient than the correct parametric formula, and it performs poorly under GARCH-type dependence because the automatic bandwidth selects too few lags Ledoit and Wolf, 2008. Further experiments traced how the convergence to the nominal level slows as the persistence of the AR and GARCH components grows. The simulations also reproduced empirically the theoretical result of Kao et al., 2025: when the fourth moment does not exist, the standard $\sqrt{T}$ scaling breaks down.

The third corrected and extended the finite-sample bias results. The widely cited approximation of Christie, 2005 rests on the false claim that $\hat{\sigma}$ is unbiased; the correct first-order factors are $1 + 3/(4T)$ with the Bessel-corrected denominator and $1 + 5/(4T)$ with denominator T , so the published correction is too small by a factor of 3/2 and 5/2 respectively. Writing the bias through the same long-run variance matrix S produced a general expression valid for every model in this work, and the per-model corrections collected in Table A.3 reduce the bias in each case considered.

Applied to three weekly SPDR series in Section 5, the hierarchical specification procedure selected an AR(1)-GARCH(1,1) model for all of them, yet the probabilistic Sharpe ratio moved only modestly across data-generating processes and not monotonically in model complexity, with one series sitting close to the conventional significance boundary.

A standard criticism of the Sharpe ratio is that, as a point estimate, it summarises performance through the first two moments alone. The inferential view taken here answers part of that objection. Section 6 showed how the third moment enters the analysis intrinsically through the asymptotic variance, and provided an introduction to the joint analysis with the Value-at-Risk.

Future work: The conditional-variance dynamics studied here are symmetric, so they ignore the leverage effect through which negative shocks raise future volatility more than positive ones; carrying the long-run variance matrix S through an asymmetric specification such as GJR-GARCH or EGARCH would give inference that respects this feature of equity returns. The analysis has also treated a single strategy at a time, whereas most decisions compare two; deriving the joint asymptotic distribution of two Sharpe ratios within the present scheme would turn the framework into a formal test of whether one strategy outperforms another. A third direction follows the line of Kao et al., 2025: a more robust and extensive study of the rate of convergence and of the irreducible bias that heavy-tailed distributions introduce.

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A Anexo

A.1 Coverage simulation methodology

As shown in Section 2.1,

$$\sqrt{T}(\widehat{SR} - SR) \xrightarrow{d} \mathcal{N}(0, V_{\widehat{SR}}), \quad (47)$$

where SR denotes the true population Sharpe ratio and $V_{\widehat{SR}}$ the asymptotic variance of the estimator, whose expression depends on the model assumed for the returns (see Section 2.2). Standardizing,

$$Z = \frac{\widehat{SR} - SR_0}{\sqrt{V_{\widehat{SR}}/T}}, \quad (48)$$

so that the symmetric, minimum-length confidence interval at level $1 - \alpha$ is

$$IC_{1-\alpha} = \left[\widehat{SR} - z_{1-\frac{\alpha}{2}} \sqrt{\frac{V_{\widehat{SR}}}{T}}, \quad \widehat{SR} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{V_{\widehat{SR}}}{T}} \right], \quad (49)$$

where $z_{1-\frac{\alpha}{2}}$ is the $1 - \alpha/2$ quantile of the $\mathcal{N}(0, 1)$ distribution.

The experimental protocol, inspired by Opdyke, 2007 but extended to multiple non-IID DGPs, is as follows. Each *configuration* is defined as a pair (*assumed model*, *DGP*). For each configuration we run n Monte Carlo iterations of the following procedure:

1. **Data generation.** A sample path of size T , $\{r_t\}_{t=1}^T$, is simulated from the chosen DGP, whose population Sharpe ratio SR is known by construction.
2. **Point estimation.** The sample Sharpe ratio is computed as $\widehat{SR} = \bar{r}/\hat{\sigma}$, using the Bessel-corrected standard deviation.
3. **Asymptotic variance estimation.** The asymptotic variance $V_{\widehat{SR}}$ is computed according to the assumptions of the postulated model. It depends on $\widehat{SR}$ and on a vector of additional parameters θ (kurtosis, skewness, autocorrelations, GARCH coefficients, etc.). We consider three alternative ways of obtaining $V_{\widehat{SR}}$:
 - **Sample moments:** $V_{\widehat{SR}} = V_{\widehat{SR}}(\widehat{SR}, \hat{\theta})$, with $\hat{\theta}$ estimated from the sample. This is the realistic scenario, as it replicates what an analyst does in practice⁵.
 - **Theoretical moments:** $V_{\widehat{SR}} = V_{\widehat{SR}}(\widehat{SR}, \theta_{DGP})$, plugging in the population parameters of the DGP. This isolates the behaviour of the asymptotic formula from the noise introduced by the estimation of θ .
 - **Non-parametric:** $V_{\widehat{SR}} = \hat{\Sigma}_{HAC}$, where $\hat{\Sigma}_{HAC}$ is the Heteroskedasticity and Autocorrelation Consistent estimator, described in Section 3.3.
4. **Interval construction.** $IC_{1-\alpha}$ is built according to (49).
5. **Evaluation.** The indicator $\mathbb{1}_{\{SR \in IC_{1-\alpha}\}}$ is recorded.

The **empirical coverage rate** is then defined as

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{SR \in IC_{1-\alpha}^{(i)}\}}. \quad (50)$$

Under correct model specification the expected value is $\mathbb{E}[\hat{p}] = 1 - \alpha$. The speed of convergence toward this target depends on the sample size T and on several other factors that the scenarios below try to showcase.

⁵Perfomed using constrained MLE estimation.

Acceptance band for coverage

To decide whether an observed empirical coverage $\hat{p}$ is statistically compatible with the nominal level $1 - \alpha$, we test

$$H_0 : p = p_0 \quad \text{vs.} \quad H_1 : p \neq p_0. \quad (51)$$

The random variable $X = \sum_{i=1}^n \mathbb{1}_{\{SR \in IC_{1-\alpha}^{(i)}\}}$ follows a binomial distribution, $X \sim \text{Bin}(n, p)$, with mean $\mathbb{E}[X] = np$ and variance $\text{Var}(X) = np(1 - p)$. Our estimator is $\hat{p} = X/n$, so $\mathbb{E}[\hat{p}] = p$ and $\text{Var}(\hat{p}) = p(1 - p)/n$. Under H_0 , by the Central Limit Theorem,

$$\frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \xrightarrow{d} \mathcal{N}(0, 1), \quad (52)$$

which defines the acceptance region of (51) at level β :

$$\hat{p} \in \left[p_0 - z_{1-\frac{\beta}{2}} \sqrt{\frac{p_0(1 - p_0)}{n}}, \quad p_0 + z_{1-\frac{\beta}{2}} \sqrt{\frac{p_0(1 - p_0)}{n}} \right]. \quad (53)$$

In our case p_0 is replaced by $1 - \alpha$. The width of this band depends only on the number of Monte Carlo replications n and on the nominal level α ; it does not depend on the sample size T , on the distributional assumptions of the DGP, or on the variance formula used. It is the band drawn around the target coverage $1 - \alpha = 0.95$ in all line plots that follow. A model is judged to have correct coverage whenever $\hat{p}$ falls inside this band.

A.2 Tail Index

The tail index $\kappa > 0$ measures how heavy the right tail of a distribution is when that tail decays like a power law. Formally, a positive random variable X has tail index κ if

$$P(X > x) = L(x) x^{-\kappa}, \quad x \rightarrow \infty, \quad (54)$$

where L is a slowly varying function, that is, $L(tx)/L(x) \rightarrow 1$ for every $t > 0$. Distributions of this type are called *regularly varying with index $-\kappa$* (Resnick et al., 1987)

Two properties make κ the natural summary of tail heaviness:

It controls which moments exist. $E|X|^p < \infty$ for $p < \kappa$ and $E|X|^p = \infty$ for $p > \kappa$. So $\kappa = 2$ is the boundary of finite variance, $\kappa = 4$ the boundary of finite fourth moment, and so on.

Smaller κ means heavier tail. A Gaussian random variable has $\kappa = \infty$. A Cauchy random variable has $\kappa = 1$.

A.2.1 The Tail Index of an AR(1)-GARCH(1,1) Process

Our work concerns AR(1)-GARCH(1,1) processes, so we now examine how the model parameters affect the tail index. Mikosch et al., 2000, Theorem 2.1(a), shows that the tail index κ of a stationary GARCH(1,1) is the unique positive solution of

$$E\left[(\alpha Z^2 + \beta)^{\kappa/2}\right] = 1, \quad (55)$$

where Z has the distribution of the innovations. A generalization to GARCH(p,q) processes is provided by Basrak et al., 2002.

The extension to AR-GARCH models is given in Lange, 2011, Theorem 1, which proves that an AR-GARCH process inherits the same tail index κ as its GARCH innovation sequence. The AR layer therefore does not alter the tail heaviness: the index is fully determined by the GARCH parameters and the distribution of Z , and the autoregressive coefficients do not enter equation (55).

To compute κ numerically, we solve (55) for κ given (α, β) . Under standard normal innovations $z_t \sim \mathcal{N}(0, 1)$, the equation reduces to finding the root of

$$\int_{-\infty}^{\infty} (\alpha z^2 + \beta)^{\kappa/2} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 1. \quad (56)$$

Table A.1 reports the solutions for the parameter pairs (α, β) employed in our simulation study 3.2.

Table A.1: Theoretical tail index κ of a GARCH(1,1) process with standard normal innovations, obtained as the numerical root of equation (56).

α	β	$\alpha + \beta$	κ
0.070	0.630	0.700	26.46
0.080	0.720	0.800	20.44
0.090	0.810	0.900	14.08
0.092	0.828	0.920	12.49
0.095	0.855	0.950	9.64
0.098	0.882	0.980	5.74

The pattern in Table A.1 reflects a well-known property of the GARCH(1,1) model: as $\alpha + \beta \rightarrow 1$, the process approaches the integrated GARCH (IGARCH) boundary and $\kappa \rightarrow 2$, so that the theoretical variance ceases to exist. Parameter configurations close to integration thus generate extremely heavy-tailed processes, while configurations far from integration produce nearly light-tailed behavior.

A.2.2 The Tail Index of an i.i.d. Student- t Distribution

For the i.i.d. case with Student- t distribution with ν degrees of freedom innovations. Its density is

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu} \Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}. \quad (57)$$

For large $|x|$, the bracketed factor admits a simple asymptotic expansion. Writing

$$\left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2} = \left(\frac{x^2}{\nu}\right)^{-(\nu+1)/2} \left(1 + \frac{\nu}{x^2}\right)^{-(\nu+1)/2}, \quad (58)$$

and noting that $(1 + \nu/x^2)^{-(\nu+1)/2} \rightarrow 1$ as $|x| \rightarrow \infty$, we obtain

$$\left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2} \sim \left(\frac{x^2}{\nu}\right)^{-(\nu+1)/2} = \nu^{(\nu+1)/2} |x|^{-(\nu+1)}. \quad (59)$$

Substituting (59) into (57) gives

$$f(x) \sim c_\nu |x|^{-(\nu+1)} \implies P(|X| > x) \sim \frac{c_\nu}{\nu} x^{-\nu}, \quad x \rightarrow \infty, \quad (60)$$

with

$$c_\nu = \frac{\Gamma(\frac{\nu+1}{2}) \nu^{\nu/2}}{\sqrt{\pi} \Gamma(\nu/2)}. \quad (61)$$

The function decays as a power law with exponent ν , which by (54) is the definition of tail index ν .

A.3 Moments of $\hat{\sigma}$

Assume that $\{X_t\}_{t=1}^T$ is an iid sample with $X_t \sim \mathcal{N}(\mu, \sigma^2)$, where both μ and σ^2 are unknown. Define

$$\bar{X} = \frac{1}{T} \sum_{t=1}^T X_t, \quad S = \sum_{t=1}^T (X_t - \bar{X})^2, \quad (62)$$

and the two usual estimators of σ^2 ,

$$\hat{\sigma}_T^2 = \frac{S}{T}, \quad \hat{\sigma}_{T-1}^2 = \frac{S}{T-1}. \quad (63)$$

The key tool is the identity $S/\sigma^2 \sim \chi_{T-1}^2$, valid under normality, from which $\mathbb{E}[S] = (T-1)\sigma^2$ and $\text{Var}(S) = 2(T-1)\sigma^4$ follow.

Moments of $\hat{\sigma}^2$. Applying the identities above:

- **Denominator $T-1$.** Unbiased,

$$\mathbb{E}[\hat{\sigma}_{T-1}^2] = \sigma^2, \quad \text{Var}(\hat{\sigma}_{T-1}^2) = \frac{2\sigma^4}{T-1}.$$

- **Denominator T .** Biased,

$$\mathbb{E}[\hat{\sigma}_T^2] = \frac{T-1}{T}\sigma^2, \quad \text{Bias}(\hat{\sigma}_T^2) = -\frac{\sigma^2}{T}, \quad \text{Var}(\hat{\sigma}_T^2) = \frac{2(T-1)\sigma^4}{T^2}.$$

The variance difference is

$$\text{Var}(\hat{\sigma}_{T-1}^2) - \text{Var}(\hat{\sigma}_T^2) = \frac{2\sigma^4(2T-1)}{T^2(T-1)} = O(T^{-2}),$$

so the choice of denominator is irrelevant to first order in T for the variance. The bias, by contrast, is $O(T^{-1})$ in the $\hat{\sigma}_T^2$ case.

Variance of $\hat{\sigma}$. To first order both estimators share the same asymptotic distribution, $\sqrt{T}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} \mathcal{N}(0, 2\sigma^4)$ (the bias of $\hat{\sigma}_T^2$ vanishes asymptotically). Applying the delta method to $f(v) = v^{1/2}$, with $f'(v) = \frac{1}{2}v^{-1/2}$:

$$\text{Var}(\hat{\sigma}) \approx [f'(\sigma^2)]^2 \text{Var}(\hat{\sigma}^2) = \frac{\sigma^2}{2T} + O(T^{-2}), \quad (64)$$

with the same first-order expression for both $\hat{\sigma}_T$ and $\hat{\sigma}_{T-1}$.

Bias of $\hat{\sigma}$. The function $f(v) = v^{1/2}$ has second derivative $f''(v) = -\frac{1}{4}v^{-3/2}$. The Taylor expansion of $\hat{\sigma} = f(\hat{\sigma}^2)$ around σ^2 is

$$\hat{\sigma} \approx \sigma + \frac{1}{2\sigma}(\hat{\sigma}^2 - \sigma^2) - \frac{1}{8\sigma^3}(\hat{\sigma}^2 - \sigma^2)^2, \quad (65)$$

and, taking expectation,

$$\mathbb{E}[\hat{\sigma} - \sigma] \approx \frac{1}{2\sigma} \text{Bias}(\hat{\sigma}^2) - \frac{1}{8\sigma^3} \mathbb{E}[(\hat{\sigma}^2 - \sigma^2)^2]. \quad (66)$$

We apply this formula to each case:

- **Denominator $T-1$.** The first term vanishes (unbiasedness of $\hat{\sigma}_{T-1}^2$) and the second matches the variance:

$$\mathbb{E}[\hat{\sigma}_{T-1} - \sigma] \approx -\frac{1}{8\sigma^3} \cdot \frac{2\sigma^4}{T-1} = -\frac{\sigma}{4(T-1)} = -\frac{\sigma}{4T} + O(T^{-2}).$$

- **Denominator T .** Both terms contribute. For the quadratic one,

$$\mathbb{E}[(\hat{\sigma}_T^2 - \sigma^2)^2] = \text{Var}(\hat{\sigma}_T^2) + \text{Bias}(\hat{\sigma}_T^2)^2 = \frac{2(T-1)\sigma^4}{T^2} + \frac{\sigma^4}{T^2} = \frac{(2T-1)\sigma^4}{T^2}.$$

Substituting into (66),

$$\mathbb{E}[\hat{\sigma}_T - \sigma] \approx \frac{1}{2\sigma} \left(-\frac{\sigma^2}{T} \right) - \frac{1}{8\sigma^3} \cdot \frac{(2T-1)\sigma^4}{T^2} = -\frac{(6T-1)\sigma}{8T^2} = -\frac{3\sigma}{4T} + O(T^{-2}).$$

Summary table. Table A.2 collects the moments of $\hat{\sigma}^2$ and $\hat{\sigma}$ to first order in T for both estimators.

	Denominator $T-1$	Denominator T
$\mathbb{E}[\hat{\sigma}^2]$	σ^2	$\frac{T-1}{T}\sigma^2$
$\text{Bias}(\hat{\sigma}^2)$	0	$-\frac{\sigma^2}{T}$
$\text{Var}(\hat{\sigma}^2)$	$\frac{2\sigma^4}{T-1}$	$\frac{2(T-1)\sigma^4}{T^2}$
$\text{Var}(\hat{\sigma})$ (first order)	$\frac{\sigma^2}{2T}$	$\frac{\sigma^2}{2T}$
$\mathbb{E}[\hat{\sigma}] - \sigma$ (first order)	$-\frac{\sigma}{4T}$	$-\frac{3\sigma}{4T}$

Table A.2: Moments of the σ^2 and σ estimators under iid normality.

A.4 Proof of the bias through the long-run variance

Let $\theta = (\mu, \sigma^2)^\top$ and its estimator $\hat{\theta} = (\hat{\mu}, \hat{\sigma}^2)^\top$. The Sharpe ratio is the function

$$f(\theta) = \frac{\mu}{\sqrt{\sigma^2}} = \frac{\mu}{\sigma}.$$

Let $\varepsilon = \hat{\theta} - \theta$ be the estimation error. Expanding $f(\hat{\theta})$ in a second-order Taylor series:

$$f(\hat{\theta}) \approx f(\theta) + \nabla f^\top \varepsilon + \frac{1}{2} \varepsilon^\top H_f \varepsilon, \quad (67)$$

where ∇f and H_f denote the gradient and the Hessian of f , evaluated at θ . Taking expectations:

$$E[f(\hat{\theta})] - f(\theta) \approx \nabla f^\top E[\varepsilon] + \frac{1}{2} \text{tr}(H_f E[\varepsilon \varepsilon^\top]).$$

Gradient and bias of $\hat{\sigma}^2$. For $f(\mu, \sigma^2) = \mu(\sigma^2)^{-1/2}$:

$$\nabla f = \begin{pmatrix} 1/\sigma \\ -\mu/(2\sigma^3) \end{pmatrix}.$$

The first bias term is

$$\nabla f^\top E[\varepsilon] = \frac{1}{\sigma} E[\hat{\mu} - \mu] - \frac{\mu}{2\sigma^3} E[\hat{\sigma}^2 - \sigma^2] = -\frac{\mu}{2\sigma^3} E[\hat{\sigma}^2 - \sigma^2],$$

since $\hat{\mu}$ is always unbiased. We need to compute $E[\hat{\sigma}^2] - \sigma^2$. Writing

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2 = \frac{1}{T-1} \sum_{t=1}^T [(r_t - \mu) - (\bar{r} - \mu)]^2,$$

expanding the square of the binomial and using $\sum_{t=1}^T (r_t - \mu) = T(\bar{r} - \mu)$, we get:

$$\hat{\sigma}^2 = \frac{1}{T-1} \left[\sum_{t=1}^T (r_t - \mu)^2 - T(\bar{r} - \mu)^2 \right].$$

Taking expectations:

$$E[\hat{\sigma}^2] = \frac{T}{T-1} [\sigma^2 - \text{Var}(\bar{r})].$$

By Proposition 7.5 of Hamilton, 1994, for a stationary process:

$$\lim_{T \rightarrow \infty} T \cdot \text{Var}(\bar{r}) = \sum_{j=-\infty}^{\infty} \text{Cov}(r_t, r_{t-j}) = S_{11},$$

so $\text{Var}(\bar{r}) \approx S_{11}/T$. Substituting:

$$E[\hat{\sigma}^2] = \frac{T\sigma^2 - S_{11}}{T-1}, \quad E[\hat{\sigma}^2] - \sigma^2 = \frac{\sigma^2 - S_{11}}{T-1}.$$

For the IID and GARCH cases, $S_{11} = \sigma^2$ and the bias is zero. Under serial dependence, an AR(1) for example, $S_{11} = \sigma^2 \frac{1+\phi}{1-\phi}$, so there is bias and its sign depends on the sign of ϕ .

Quadratic term: variance of the estimator. Since $\varepsilon^\top H_f \varepsilon$ is a scalar, it can be written as its own trace, and applying cyclicity:

$$E[\varepsilon^\top H_f \varepsilon] = \text{tr}(H_f E[\varepsilon \varepsilon^\top]).$$

The estimator $\hat{\theta}$ is the sample average of the moment-condition vector $m_t = (r_t - \mu, (r_t - \mu)^2 - \sigma^2)^\top$. The CLT for α -mixing processes guarantees that $\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} N(0, S)$ (as explained in the methodology section), where $S = \sum_{j=-\infty}^{\infty} E[m_t m_{t-j}^\top]$ is the same long-run variance matrix obtained in Section 2.2. Therefore:

$$E[\varepsilon \varepsilon^\top] = \text{Var}(\hat{\theta}) \approx \frac{S}{T} = \frac{1}{T} \begin{pmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{pmatrix}.$$

The Hessian of $f(\mu, \sigma^2) = \mu(\sigma^2)^{-1/2}$ is

$$H_f = \begin{pmatrix} 0 & -1/(2\sigma^3) \\ -1/(2\sigma^3) & 3\mu/(4\sigma^5) \end{pmatrix}.$$

Expanding the trace:

$$\text{tr}\left(H_f \cdot \frac{S}{T}\right) = \frac{1}{T} \left(-\frac{S_{12}}{2\sigma^3} - \frac{S_{12}}{2\sigma^3} + \frac{3\mu}{4\sigma^5} S_{22} \right) = \frac{1}{T} \left(-\frac{S_{12}}{\sigma^3} + \frac{3\mu}{4\sigma^5} S_{22} \right).$$

Final bias expression. Collecting both terms and using $\mu/\sigma = SR$:

$$\boxed{E[\widehat{SR}] - SR \approx \frac{SR}{2(T-1)} \left(\frac{S_{11}}{\sigma^2} - 1 \right) + \frac{1}{2T} \left(-\frac{S_{12}}{\sigma^3} + \frac{3SR}{4\sigma^4} S_{22} \right)}. \quad (68)$$

This expression is fully general and depends only on the elements of S , which in turn depend on the assumed model for the returns.

Solving for SR (which we will call $\widehat{SR}_{\text{corrected}}$):

$$\boxed{\widehat{SR}_{\text{corrected}} = \left(\widehat{SR} + \frac{1}{2T} \frac{S_{12}}{\sigma^3} \right) \left[1 + \frac{1}{2(T-1)} \left(\frac{S_{11}}{\sigma^2} - 1 \right) + \frac{3}{8T\sigma^4} S_{22} \right]^{-1}} \quad (69)$$

A.5 Sharpe ratio estimator corrections by model

Table A.3: Sample Sharpe ratio corrections by model ($\widehat{SR}_{\text{corrected}}$)

Model	Return equation	Correction ($\widehat{SR}_{\text{corrected}}$)
IID Normal	$r_t \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$	$\widehat{SR} \left[1 + \frac{3}{4T} \right]^{-1}$
IID Non-Normal	$r_t \stackrel{\text{iid}}{\sim} (\mu, \sigma^2)$	$\left(\widehat{SR} + \frac{\gamma}{2T} \right) \left[1 + \frac{3(\kappa - 1)}{8T} \right]^{-1}$
AR(1) Normal	$\begin{cases} r_t = \mu_0 + \phi r_{t-1} + \varepsilon_t \\ \varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2) \end{cases}$	$\widehat{SR} \left[1 + \frac{\phi}{(T-1)(1-\phi)} + \frac{3}{4T} \frac{1+\phi^2}{1-\phi^2} \right]^{-1}$
AR(1)	$\begin{cases} r_t = \mu_0 + \phi r_{t-1} + \varepsilon_t \\ \varepsilon_t \stackrel{\text{iid}}{\sim} (0, \sigma_\varepsilon^2) \end{cases}$	$\left(\widehat{SR} + \frac{\gamma}{2T} \frac{1+\phi+\phi^2}{1-\phi^2} \right) \left[1 + \frac{\phi}{(T-1)(1-\phi)} + \frac{3(\kappa-1)}{8T} \frac{1+\phi^2}{1-\phi^2} \right]^{-1}$
GARCH(1,1)	$\begin{cases} r_t = \mu + u_t, u_t = \sigma_t \varepsilon_t \\ \sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2 \end{cases}$	$\left(\widehat{SR} + \frac{\gamma}{2T} \frac{1-\beta}{(1-\alpha-\beta)} \right) \left[1 + \frac{3(\kappa-1)}{8T} \frac{(1-\beta)^2(1+\alpha+\beta)}{(1-\alpha-\beta)(1-2\alpha\beta-\beta^2)} \right]^{-1}$

Note: $\gamma = E[(r_t - \mu)^3]/\sigma^3$ denotes the skewness and $\kappa = E[(r_t - \mu)^4]/\sigma^4$ the kurtosis of r_t . T is the sample size.

A.6 Model selection Diagram

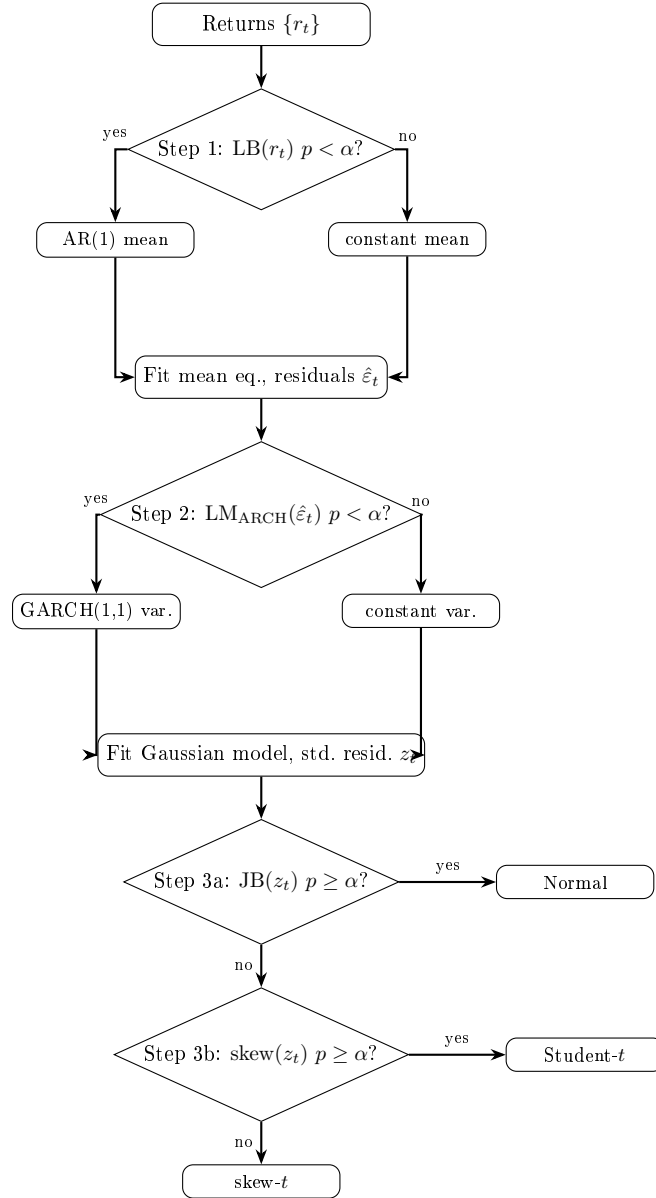


Figure A.1: Hierarchical specification procedure.

B Asymptotic Variance of the Sharpe Ratio Estimator

This appendix derives the asymptotic variance of the Sharpe ratio estimator $\widehat{SR}$ when returns follow an AR(1)-GARCH(1,1) process. First, we explain some time series theory that is used across the solution; then we state some preliminary results and finally we find the entries of the long-run variance matrix S , namely S_{11} , S_{12} , S_{22} , and apply the delta method to obtain the asymptotic variance. This follows the theory developed in Section 2.1.

B.1 Linear filters

The derivation rests on a standard filtering identity. The treatment follows Hamilton, 1994, Ch. 3 for lag operators and Wold form, Ch. 6 for the autocovariance-generating function, and Ch. 10 for the vector case.

Univariate case. Let y_t be a covariance-stationary scalar process with absolutely summable Wold representation

$$y_t = \psi(L) \varepsilon_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \sum_{j=0}^{\infty} |\psi_j| < \infty,$$

where ε_t is mean-zero white noise with variance σ_ε^2 and $\psi(L) = \sum_j \psi_j L^j$. The autocovariance-generating function of y_t (Hamilton Ch. 6) packages all autocovariances $\gamma_h \equiv E[y_t y_{t-h}]$ into a single object,

$$g_y(z) \equiv \sum_{h=-\infty}^{\infty} \gamma_h z^h = \sigma_\varepsilon^2 \psi(z) \psi(z^{-1}).$$

The long-run variance is obtained by evaluating g_y at $z = 1$:

$$S \equiv \sum_{h=-\infty}^{\infty} \gamma_h = g_y(1) = \sigma_\varepsilon^2 \psi(1)^2.$$

Multivariate case. Let $y_t \in \mathbb{R}^n$ be vector covariance-stationary with representation

$$y_t = \Psi(L) \varepsilon_t = \sum_{j=0}^{\infty} \Psi_j \varepsilon_{t-j}, \quad \sum_{j=0}^{\infty} \|\Psi_j\| < \infty,$$

where $\varepsilon_t \in \mathbb{R}^m$ is mean-zero white noise with covariance $\Sigma_\varepsilon = E[\varepsilon_t \varepsilon_t^\top]$ and $\Psi(L)$ is an $n \times m$ matrix polynomial in the lag operator. The vector analogue of the autocovariance-generating function (Hamilton Ch. 10, spectral representation of multivariate processes) is

$$G_y(z) \equiv \sum_{h=-\infty}^{\infty} \Gamma_h z^h = \Psi(z) \Sigma_\varepsilon \Psi(z^{-1})^\top,$$

where $\Gamma_h \equiv E[y_t y_{t-h}^\top]$. Evaluating at $z = 1$ yields the long-run covariance matrix

$$S = \sum_{h=-\infty}^{\infty} \Gamma_h = \Psi(1) \Sigma_\varepsilon \Psi(1)^\top. \quad (\text{LR})$$

The (i, k) entry of S is the long-run covariance between the i -th and k -th components of y_t and is obtained by sandwiching Σ_ε between the i -th row of $\Psi(1)$ and the k -th column of $\Psi(1)^\top$. If ε_t is a martingale difference sequence, the cross-lag autocovariances of ε_t are exactly zero, so Σ_ε is precisely the contemporaneous covariance.

B.2 Preliminary results

Consider the AR(1)–GARCH(1,1) model

$$r_t = c + \phi r_{t-1} + u_t, \quad u_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2,$$

with $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} (0, 1)$, skewness skew_ϵ and kurtosis k_ϵ . Let $x_t \equiv r_t - c/(1 - \phi)$ denote the centered return, so that $x_t = \phi x_{t-1} + u_t$. Throughout we assume covariance stationarity: $|\phi| < 1$ and $\alpha + \beta < 1$.

Unconditional variances. Taking expectations in the GARCH recursion gives the unconditional variance of u_t :

$$\sigma_u^2 \equiv E[u_t^2] = \frac{\omega}{1 - \alpha - \beta}. \quad (\text{PR1})$$

Since $x_t = \phi x_{t-1} + u_t$ has a white-noise innovation, stationarity yields

$$\sigma^2 \equiv E[x_t^2] = \frac{\sigma_u^2}{1 - \phi^2}. \quad (\text{PR2})$$

An MDS innovation vector. Define

$$e_t = \begin{pmatrix} u_t \\ \xi_t \\ \eta_t \end{pmatrix} \equiv \begin{pmatrix} u_t \\ x_{t-1}u_t \\ u_t^2 - \sigma_t^2 \end{pmatrix}. \quad (\text{PR3})$$

Let $\mathcal{F}_{t-1} = \sigma(u_{t-1}, u_{t-2}, \dots)$. Both σ_t^2 and x_{t-1} are $\mathcal{F}_{t-1}$ -measurable, so

$$\begin{aligned} E[u_t | \mathcal{F}_{t-1}] &= \sigma_t E[\epsilon_t] = 0, \\ E[\xi_t | \mathcal{F}_{t-1}] &= x_{t-1} E[u_t | \mathcal{F}_{t-1}] = 0, \\ E[\eta_t | \mathcal{F}_{t-1}] &= \sigma_t^2 E[\epsilon_t^2] - \sigma_t^2 = 0. \end{aligned}$$

Hence e_t is a martingale difference sequence, and by iterated expectations $E[e_t e_{t-h}^\top] = 0$ for every $h \neq 0$. Consequently the long-run innovation covariance collapses to the contemporaneous one,

$$S_{ee} \equiv \sum_{h=-\infty}^{\infty} E[e_t e_{t-h}^\top] = E[e_t e_t^\top] \equiv \Sigma_{ee}. \quad (\text{PR4})$$

Third-moment cross terms of Σ_{ee} . Since $E[u_t \sigma_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2 E[u_t | \mathcal{F}_{t-1}] = 0$,

$$\Sigma_{u\eta} \equiv E[u_t \eta_t] = E[u_t^3] - E[u_t \sigma_t^2] = E[u_t^3]. \quad (\text{PR5})$$

For the other cross term, conditioning on $\mathcal{F}_{t-1}$ gives $\Sigma_{u\xi} = E[x_{t-1} u_t^2] = E[x_{t-1} \sigma_t^2]$. Substituting the GARCH recursion and using $E[x_{t-1}] = 0$,

$$E[x_{t-1} \sigma_t^2] = \alpha E[x_{t-1} u_{t-1}^2] + \beta E[x_{t-1} \sigma_{t-1}^2].$$

With $x_{t-1} = \phi x_{t-2} + u_{t-1}$, stationarity (so $E[x_{t-2} \sigma_{t-1}^2] = E[x_{t-1} \sigma_t^2]$), and $E[u_{t-1} \sigma_{t-1}^2] = 0$,

$$\begin{aligned} E[x_{t-1} u_{t-1}^2] &= \phi E[x_{t-1} \sigma_t^2] + E[u_t^3], \\ E[x_{t-1} \sigma_{t-1}^2] &= \phi E[x_{t-1} \sigma_t^2]. \end{aligned}$$

Setting $A \equiv E[x_{t-1} \sigma_t^2]$, $A = \alpha(\phi A + E[u_t^3]) + \beta \phi A = \phi(\alpha + \beta)A + \alpha E[u_t^3]$, so

$$\Sigma_{u\xi} = \frac{\alpha E[u_t^3]}{1 - \phi(\alpha + \beta)}. \quad (\text{PR6})$$

The third moment of u_t in terms of skew_r . Cubing $x_t = \phi x_{t-1} + u_t$,

$$x_t^3 = \phi^3 x_{t-1}^3 + 3\phi^2 x_{t-1}^2 u_t + 3\phi x_{t-1} u_t^2 + u_t^3.$$

The MDS property kills $E[x_{t-1}^2 u_t] = E[x_{t-1}^2 E[u_t | \mathcal{F}_{t-1}]] = 0$, and stationarity gives $(1 - \phi^3)E[x_t^3] = 3\phi \Sigma_{u\xi} + E[u_t^3]$. Inserting (PR6),

$$(1 - \phi^3) E[x_t^3] = E[u_t^3] \left[\frac{3\phi\alpha}{1 - \phi(\alpha + \beta)} + 1 \right] = E[u_t^3] \frac{1 + 2\phi\alpha - \phi\beta}{1 - \phi(\alpha + \beta)}.$$

Since $E[x_t^3] = \sigma^3 \text{skew}_r$,

$$E[u_t^3] = \sigma^3 \text{skew}_r (1 - \phi^3) \frac{1 - \phi(\alpha + \beta)}{1 + 2\phi\alpha - \phi\beta}. \quad (\text{PR7})$$

ARMA(1,1) representation of $u_t^2 - \sigma_u^2$. Writing $u_t^2 - \sigma_u^2 = (\sigma_t^2 - \sigma_u^2) + \eta_t$ and noting that the GARCH recursion in deviation form reads $(1 - \beta L)(\sigma_t^2 - \sigma_u^2) = \alpha L(u_t^2 - \sigma_u^2)$, elimination of $\sigma_t^2 - \sigma_u^2$ gives the ARMA(1,1) representation

$$(1 - (\alpha + \beta)L)(u_t^2 - \sigma_u^2) = (1 - \beta L)\eta_t, \quad u_t^2 - \sigma_u^2 = \frac{1 - \beta L}{1 - (\alpha + \beta)L} \eta_t. \quad (\text{PR8})$$

This representation will be reused both when constructing the variance filter and when computing $\Sigma_{\eta\eta}$.

Autocovariance of u_t^2 . Decompose

$$E[u_t^2 u_{t+k}^2] = \gamma_k(u^2) + \sigma_u^4, \quad \gamma_k(u^2) \equiv \text{Cov}(u_t^2, u_{t+k}^2).$$

From the ARMA(1,1) form (PR8), $\gamma_k(u^2) = (\alpha + \beta)^{k-1} \gamma_1(u^2)$ for $k \geq 1$, with first autocovariance (Hamilton, Ch. 3)

$$\gamma_1(u^2) = (k_u - 1) \sigma_u^4 \frac{\alpha(1 - \alpha\beta - \beta^2)}{1 - 2\alpha\beta - \beta^2}, \quad (\text{PR9})$$

where $k_u \equiv E[u_t^4]/\sigma_u^4$ is the kurtosis of u_t .

Variance of η_t : $\Sigma_{\eta\eta}$. The unconditional variance of an ARMA(1,1) process $(1 - \phi L)y_t = (1 + \theta L)\nu_t$ is $\sigma_\nu^2(1 + 2\phi\theta + \theta^2)/(1 - \phi^2)$. Applying this to (PR8) with $\phi = \alpha + \beta$, $\theta = -\beta$, $\text{Var}(u_t^2 - \sigma_u^2) = (k_u - 1)\sigma_u^4$, and innovation variance $\sigma_\eta^2 = \Sigma_{\eta\eta}$ gives $(k_u - 1)\sigma_u^4 = \frac{1 - 2\alpha\beta - \beta^2}{1 - (\alpha + \beta)^2} \Sigma_{\eta\eta}$, i.e.,

$$\Sigma_{\eta\eta} = (k_u - 1) \sigma_u^4 \frac{1 - (\alpha + \beta)^2}{1 - 2\alpha\beta - \beta^2}. \quad (\text{PR10})$$

$\Sigma_{\xi\eta}$ and the cross moment M_1 . Define

$$M_1 \equiv E[x_{t-1} u_t^3]. \quad (\text{M1})$$

By definition, $\Sigma_{\xi\eta} = E[x_{t-1} u_t(u_t^2 - \sigma_t^2)] = E[x_{t-1} u_t^3] - E[x_{t-1} u_t \sigma_t^2]$. The second term vanishes because $x_{t-1} \sigma_t^2$ is $\mathcal{F}_{t-1}$ -measurable and $E[u_t | \mathcal{F}_{t-1}] = 0$, so

$$\Sigma_{\xi\eta} = M_1. \quad (\text{PR11})$$

Recursion for $\Sigma_{\xi\xi}$. By stationarity it is convenient to define

$$H_k \equiv E[x_t^2 u_{t+k}^2], \quad k \geq 1,$$

so that $\Sigma_{\xi\xi} = \text{Var}(x_{t-1} u_t) = E[x_{t-1}^2 u_t^2] = H_1$. Substituting $x_t^2 = \phi^2 x_{t-1}^2 + 2\phi x_{t-1} u_t + u_t^2$ and using stationarity ($E[x_{t-1}^2 u_{t+k}^2] = H_{k+1}$),

$$H_k = \phi^2 H_{k+1} + 2\phi E[x_{t-1} u_t u_{t+k}^2] + E[u_t^2 u_{t+k}^2].$$

The cross term admits a clean reduction. Conditioning on $\mathcal{F}_{t+k-1}$ and applying the GARCH recursion,

$$E[x_{t-1} u_t u_{t+k}^2] = \alpha E[x_{t-1} u_t u_{t+k-1}^2] + \beta E[x_{t-1} u_t \sigma_{t+k-1}^2],$$

using $E[x_{t-1} u_t] = 0$. For $k \geq 2$, write $\sigma_{t+k-1}^2 = u_{t+k-1}^2 - \eta_{t+k-1}$ and note that $x_{t-1} u_t$ is $\mathcal{F}_{t+k-2}$ -measurable while $E[\eta_{t+k-1} | \mathcal{F}_{t+k-2}] = 0$, so $E[x_{t-1} u_t \eta_{t+k-1}] = 0$ by iterated expectations. Hence $E[x_{t-1} u_t \sigma_{t+k-1}^2] = E[x_{t-1} u_t u_{t+k-1}^2]$, giving the geometric recursion

$$E[x_{t-1} u_t u_{t+k}^2] = (\alpha + \beta) E[x_{t-1} u_t u_{t+k-1}^2], \quad k \geq 2.$$

For $k = 1$, $E[x_{t-1} u_t \sigma_t^2] = 0$ by the MDS argument used in (PR6), leaving $E[x_{t-1} u_t u_{t+1}^2] = \alpha E[x_{t-1} u_t^3] = \alpha M_1$. Therefore

$$E[x_{t-1} u_t u_{t+k}^2] = \alpha(\alpha + \beta)^{k-1} M_1, \quad k \geq 1.$$

Combining this with (PR9),

$$H_k = \phi^2 H_{k+1} + 2\phi\alpha(\alpha + \beta)^{k-1} M_1 + (\alpha + \beta)^{k-1} \gamma_1(u^2) + \sigma_u^4.$$

Iterating from H_1 ,

$$H_1 = \phi^{2k} H_{k+1} + (2\phi\alpha M_1 + \gamma_1(u^2)) \sum_{j=1}^k \phi^{2(j-1)} (\alpha + \beta)^{j-1} + \sigma_u^4 \sum_{j=1}^k \phi^{2(j-1)}.$$

Under $|\phi| < 1$ and $\alpha + \beta < 1$, $\phi^{2k} H_{k+1} \rightarrow 0$, and both geometric series converge:

$$\Sigma_{\xi\xi} = H_1 = \frac{1}{1 - \phi^2(\alpha + \beta)} \left[2\phi\alpha M_1 + (k_u - 1) \sigma_u^4 \frac{\alpha(1 - \alpha\beta - \beta^2)}{1 - 2\alpha\beta - \beta^2} \right] + \frac{\sigma_u^4}{1 - \phi^2}. \quad (\text{PR12})$$

B.3 S_{11}

The mean moment condition is x_t itself. From $(1 - \phi L)x_t = u_t$,

$$x_t = \frac{1}{1 - \phi L} u_t = H_x(L) e_t, \quad H_x(L) = \begin{pmatrix} \frac{1}{1 - \phi L} & 0 & 0 \end{pmatrix},$$

so $H_x(1) = (1/(1 - \phi) \ 0 \ 0)$. By (LR) and (PR4),

$$S_{11} = H_x(1) \Sigma_{ee} H_x(1)^\top = \frac{\sigma_u^2}{(1 - \phi)^2}.$$

Substituting (PR1) and (PR2), which together yield $\sigma_u^2 = (1 - \phi^2)\sigma^2 = (1 - \phi)(1 + \phi)\sigma^2$, gives

$$S_{11} = \frac{1 + \phi}{1 - \phi} \sigma^2.$$

B.4 S_{12}

Joint filter setup. Collect the bivariate moment process $w_t = (x_t, x_t^2 - \sigma^2)^\top$ and write it as a linear filter of the three-dimensional MDS innovation e_t from (PR3):

$$w_t = H(L) e_t, \quad H(L) = \begin{pmatrix} H_x(L) \\ H_{x^2}(L) \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

By (LR) and (PR4), the bivariate long-run variance is $S = H(1) \Sigma_{ee} H(1)^\top$, whose (1,2) entry reads

$$S_{12} = H_x(1) \Sigma_{ee} H_{x^2}(1)^\top.$$

$H_x(L)$ is already constructed in the S_{11} derivation; we now build $H_{x^2}(L)$.

Variance filter. Squaring $x_t = \phi x_{t-1} + u_t$ and subtracting σ^2 , using (PR2),

$$(1 - \phi^2 L)(x_t^2 - \sigma^2) = 2\phi \xi_t + (u_t^2 - \sigma_u^2).$$

Substituting the ARMA(1,1) form (PR8),

$$x_t^2 - \sigma^2 = \frac{2\phi}{1 - \phi^2 L} \xi_t + \frac{1 - \beta L}{(1 - \phi^2 L)(1 - (\alpha + \beta)L)} \eta_t = H_{x^2}(L) e_t,$$

with

$$H_{x^2}(L) = \begin{pmatrix} 0 & \frac{2\phi}{1 - \phi^2 L} & \frac{1 - \beta L}{(1 - \phi^2 L)(1 - (\alpha + \beta)L)} \end{pmatrix}.$$

Matrix expression. Evaluating at $L = 1$,

$$H_{x^2}(1)^\top = \begin{pmatrix} 0 \\ \frac{2\phi}{1 - \phi^2} \\ \frac{1 - \beta}{(1 - \phi^2)(1 - \alpha - \beta)} \end{pmatrix}.$$

Partitioning

$$\Sigma_{ee} = \begin{pmatrix} \sigma_u^2 & \Sigma_{u\xi} & \Sigma_{u\eta} \\ \Sigma_{u\xi} & \Sigma_{\xi\xi} & \Sigma_{\xi\eta} \\ \Sigma_{u\eta} & \Sigma_{\xi\eta} & \Sigma_{\eta\eta} \end{pmatrix},$$

the row-times-matrix product picks up only the top row,

$$H_x(1) \Sigma_{ee} = \frac{1}{1-\phi} \begin{pmatrix} \sigma_u^2 & \Sigma_{u\xi} & \Sigma_{u\eta} \end{pmatrix},$$

and the first component drops out when contracting with $H_{x^2}(1)^\top$, leaving

$$S_{12} = \frac{2\phi \Sigma_{u\xi}}{(1-\phi)(1-\phi^2)} + \frac{(1-\beta) \Sigma_{u\eta}}{(1-\phi)(1-\phi^2)(1-\alpha-\beta)}.$$

Solution. Inserting (PR5) and (PR6),

$$S_{12} = \frac{E[u_t^3]}{(1-\phi)(1-\phi^2)} \left[\frac{2\phi\alpha}{1-\phi(\alpha+\beta)} + \frac{1-\beta}{1-\alpha-\beta} \right].$$

Substituting (PR7) and cancelling $(1-\phi)$ via $1-\phi^3 = (1-\phi)(1+\phi+\phi^2)$,

$$S_{12} = \frac{\sigma^3 \text{skew}_r (1+\phi+\phi^2)}{(1-\phi^2)(1+2\phi\alpha-\phi\beta)} \left[2\phi\alpha + \frac{(1-\beta)(1-\phi(\alpha+\beta))}{1-\alpha-\beta} \right].$$

S_{12} is proportional to skew_r and therefore vanishes under symmetric returns. Consistency checks: (i) when $\phi = 0$, the bracket collapses to $(1-\beta)/(1-\alpha-\beta)$ and $S_{12} = \sigma^3 \text{skew}_r (1-\beta)/(1-\alpha-\beta)$, the pure-GARCH case; (ii) when $\alpha = \beta = 0$, the bracket equals 1, recovering $S_{12} = \sigma^3 \text{skew}_r (1+\phi+\phi^2)/(1-\phi^2)$.

B.5 S_{22}

The variance moment condition is $x_t^2 - \sigma^2 = H_{x^2}(L) e_t$, so by the same logic as for S_{12} ,

$$S_{22} = H_{x^2}(1) \Sigma_{ee} H_{x^2}(1)^\top.$$

Reading off the filter from the S_{12} subsection, $H_{x^2}(1) = \begin{pmatrix} 0 & A & B \end{pmatrix}$, where

$$A \equiv \frac{2\phi}{1-\phi^2}, \quad B \equiv \frac{1-\beta}{(1-\phi^2)(1-\alpha-\beta)}. \quad (\text{AB})$$

Since $H_{x^2}(1)$ has a zero in its first slot, only the lower-right 2×2 block of Σ_{ee} contributes, and

$$S_{22} = A^2 \Sigma_{\xi\xi} + 2AB \Sigma_{\xi\eta} + B^2 \Sigma_{\eta\eta}. \quad (\text{S22})$$

The three required entries of Σ_{ee} have been computed in the preliminary results: $\Sigma_{\xi\xi}$ in (PR12), $\Sigma_{\eta\eta}$ in (PR10), and $\Sigma_{\xi\eta} = M_1$ in (PR11). Substituting,

$$S_{22} = A^2 \left\{ \frac{2\phi\alpha M_1 + (k_u - 1)\sigma_u^4 \alpha \frac{1-\alpha\beta-\beta^2}{1-2\alpha\beta-\beta^2}}{1-\phi^2(\alpha+\beta)} + \frac{\sigma_u^4}{1-\phi^2} \right\} + 2AB M_1 + B^2 (k_u - 1)\sigma_u^4 \frac{1-(\alpha+\beta)^2}{1-2\alpha\beta-\beta^2}.$$

This expression depends on two ingredients that must still be expressed in terms of return moments: the cross moment $M_1 = E[x_{t-1} u_t^3]$ and the unconditional kurtosis k_u of u_t . We address each in turn.

Approximation for M_1 . Conditioning on $\mathcal{F}_{t-1}$ and exploiting $\epsilon_t \perp \mathcal{F}_{t-1}$, the cross moment factors as

$$M_1 = E[x_{t-1}\sigma_t^3\epsilon_t^3] = \text{skew}_\epsilon \cdot E[x_{t-1}\sigma_t^3].$$

The remaining expectation involves the odd half-power $\sigma_t^3 = (\sigma_t^2)^{3/2}$, which is non-polynomial in the GARCH specification and does not satisfy a closed recursion. No exact closed form is known for M_1 under a general GARCH(1,1); we therefore employ a first-order Taylor approximation.

By Taylor's theorem applied to $f(y) = y^{3/2}$ around $y = \sigma_u^2$, with $f'(\sigma_u^2) = \frac{3}{2}\sigma_u$ and $f''(y) = \frac{3}{4}y^{-1/2}$,

$$\sigma_t^3 = \sigma_u^3 + \frac{3}{2}\sigma_u(\sigma_t^2 - \sigma_u^2) + O((\sigma_t^2 - \sigma_u^2)^2).$$

Since $E[x_{t-1}] = 0$ and $E[x_{t-1}\sigma_t^2] = \Sigma_{u\xi}$ from (PR6),

$$E[x_{t-1}\sigma_t^3] \approx \frac{3}{2}\sigma_u \cdot \frac{\alpha E[u_t^3]}{1 - \phi(\alpha + \beta)}.$$

Replacing $\text{skew}_\epsilon \approx E[u_t^3]/\sigma_u^3$ in the factored expression and substituting,

$$M_1 \approx \frac{3\alpha (E[u_t^3])^2}{2\sigma_u^2 (1 - \phi(\alpha + \beta))}.$$

Plugging in (PR7) and $\sigma_u^2 = (1 - \phi^2)\sigma^2$, with the identity $(1 - \phi^3)^2/(1 - \phi^2) = (1 - \phi)(1 + \phi + \phi^2)^2/(1 + \phi)$,

$$\boxed{M_1 \approx m_1 \sigma^4 \text{skew}_r^2, \quad m_1 \equiv \frac{3\alpha (1 - \phi) (1 + \phi + \phi^2)^2 (1 - \phi(\alpha + \beta))}{2 (1 + \phi) (1 + 2\phi\alpha - \phi\beta)^2}.} \quad (\text{M1}^*)$$

Two features deserve mention. First, the approximation is exact to leading order in the GARCH excess variability $\sigma_t^2 - \sigma_u^2$ and therefore tightens as $\alpha + \beta \rightarrow 0$. Second, M_1 enters S_{22} multiplied by terms that vanish at $\phi = 0$ or $\alpha = 0$, so any approximation error is further attenuated in weakly persistent regimes.

Kurtosis k_u in terms of k_r . Squaring $x_t^2 = \phi^2 x_{t-1}^2 + 2\phi x_{t-1}u_t + u_t^2$ once more, taking expectations, and using $E[x_{t-1}^3 u_t] = 0$ from the MDS property,

$$(1 - \phi^4) E[x_t^4] = 6\phi^2 H_1 + 4\phi M_1 + k_u \sigma_u^4, \quad H_1 = \Sigma_{\xi\xi}.$$

Writing $E[x_t^4] = k_r \sigma^4$, $\sigma_u^4 = (1 - \phi^2)^2 \sigma^4$, $1 - \phi^4 = (1 - \phi^2)(1 + \phi^2)$, and introducing the convenient auxiliary

$$K \equiv (k_u - 1) \sigma_u^4,$$

this relation becomes

$$K = (1 - \phi^2) \sigma^4 [(1 + \phi^2) k_r - (1 - \phi^2)] - 6\phi^2 H_1 - 4\phi M_1.$$

Substituting H_1 from (PR12), written as

$$H_1 = \frac{2\phi\alpha M_1 + K P}{Q} + (1 - \phi^2) \sigma^4, \quad P \equiv \frac{\alpha(1 - \alpha\beta - \beta^2)}{1 - 2\alpha\beta - \beta^2}, \quad Q \equiv 1 - \phi^2(\alpha + \beta),$$

collecting K on the left-hand side, and using the simplifications $3\phi^2\alpha + Q = 1 + \phi^2(2\alpha - \beta)$ and $-(1 - \phi^2) - 6\phi^2 = -1 - 5\phi^2$,

$$\boxed{K \equiv (k_u - 1) \sigma_u^4 = \frac{Q (1 - \phi^2) \sigma^4 [(1 + \phi^2) k_r - 1 - 5\phi^2] - 4\phi [1 + \phi^2(2\alpha - \beta)] M_1}{Q + 6\phi^2 P}.} \quad (\text{K})$$

Assembly. With M_1 from (M1*) and K from (K), every ingredient of S_{22} is now a function of (ϕ, α, β) and the return moments $\{k_r, \text{skew}_r\}$. Substituting both into the pre-assembly expression for S_{22} and reorganising as a linear form in $\{1, k_r, \text{skew}_r^2\}$, after a routine of tedious algebraic steps the result takes the compact form

$$S_{22} = \sigma^4 \left[c_0 + c_k k_r + c_s \text{skew}_r^2 \right], \quad (\text{S22*})$$

with three coefficients c_0, c_k, c_s depending only on the dynamics parameters. The explicit expressions for c_0, c_k, c_s , together with all auxiliary quantities, are collected in the final-assembly section below.

Remark: symmetric innovation. When the standardised innovation is symmetric ($\text{skew}_\epsilon = 0$), both $M_1 = 0$ and $S_{12} = 0$ exactly, and (S22*) reduces to

$$S_{22}|_{\text{sym}} = \frac{\sigma^4}{1 - \phi^2} \left[4\phi^2 + \frac{\mathcal{K}}{1 + \mathcal{G}} \left(\frac{2}{3} \mathcal{G} + \frac{(1 - \beta)^2 (1 + \alpha + \beta)}{(1 - \alpha - \beta)(1 - 2\alpha\beta - \beta^2)} \right) \right],$$

where the auxiliary symmetric-case quantities are

$$\mathcal{G} \equiv \frac{6\phi^2 \alpha (1 - \alpha - \beta)}{(1 - 2\alpha\beta - \beta^2) [1 - \phi^2(\alpha + \beta)]}, \quad \mathcal{K} \equiv (1 + \phi^2) k_r - 5\phi^2 - 1.$$

B.6 Final assembly

Combining the long-run covariance entries S_{11}, S_{12}, S_{22} via the delta-method formula derived in Section 2.1,

$$V_{SR} = \frac{S_{11}}{\sigma^2} - \frac{\mu S_{12}}{\sigma^4} + \frac{\mu^2 S_{22}}{4\sigma^6}. \quad (\text{DM})$$

Substituting and factoring out $\mu/\sigma \equiv SR$, and expressing each piece in terms of the return-level moments $\{\text{skew}_r, k_r\}$,

$$V_{SR} = \frac{1 + \phi}{1 - \phi} - SR \cdot b \text{skew}_r + \frac{SR^2}{4} \left[c_0 + c_k k_r + c_s \text{skew}_r^2 \right]. \quad (\text{V})$$

Auxiliary variables. Let $\rho \equiv \alpha + \beta$. The basic building blocks are

$$P \equiv \frac{\alpha(1 - \alpha\beta - \beta^2)}{1 - 2\alpha\beta - \beta^2}, \quad Q \equiv 1 - \phi^2 \rho, \quad R \equiv \frac{1 - \rho^2}{1 - 2\alpha\beta - \beta^2},$$

together with the filter values from (AB),

$$A \equiv \frac{2\phi}{1 - \phi^2}, \quad B \equiv \frac{1 - \beta}{(1 - \phi^2)(1 - \rho)}.$$

The skewness coefficient of S_{12} :

$$b = \frac{1 + \phi + \phi^2}{(1 - \phi^2)(1 + 2\phi\alpha - \phi\beta)} \left[2\phi\alpha + \frac{(1 - \beta)(1 - \phi\rho)}{1 - \rho} \right].$$

The block coefficients of $S_{22} = \sigma^4 T_0 + T_1 M_1 + T_2 K$ are

$$\begin{aligned} T_0 &= \frac{4\phi^2}{1 - \phi^2}, \\ T_1 &= \frac{2\phi\alpha A^2}{Q} + 2AB = \frac{4\phi}{(1 - \phi^2)^2} \left[\frac{2\phi^2 \alpha}{Q} + \frac{1 - \beta}{1 - \rho} \right], \\ T_2 &= \frac{A^2 P}{Q} + B^2 R = \frac{1}{(1 - \phi^2)^2 (1 - 2\alpha\beta - \beta^2)} \left[\frac{4\phi^2 \alpha (1 - \alpha\beta - \beta^2)}{Q} + \frac{(1 - \beta)^2 (1 + \rho)}{1 - \rho} \right]. \end{aligned}$$

Finally, the S_{22} coefficients in primitives are

$$\begin{aligned}
c_0 &= T_0 - \frac{T_2 Q (1 - \phi^2) (1 + 5\phi^2)}{Q + 6\phi^2 P}, \\
c_k &= \frac{T_2 Q (1 - \phi^2) (1 + \phi^2)}{Q + 6\phi^2 P}, \\
c_s &= \frac{3\alpha (1 - \phi) (1 + \phi + \phi^2)^2 (1 - \phi\rho)}{2 (1 + \phi) (1 + 2\phi\alpha - \phi\beta)^2} \left[T_1 - \frac{4\phi [1 + \phi^2(2\alpha - \beta)] T_2}{Q + 6\phi^2 P} \right].
\end{aligned}$$